

How to actually use Claude

Real work with Claude Desktop and Claude Code

Ali Habibullah · KAUST Academy

BEFORE

A vague request gets a vague answer

What I typed:

Write an email about the safety training.

What I got:

- Three generic versions
- **5 blanks** I still have to fill in

Works on Free

Real reply, Claude Desktop (Free plan), 27 September 2026

Write an email about the safety training.

Drafting a training-related email despite missing details.

I'll draft this as a general announcement/reminder about an upcoming safety training — let me know if you meant something more specific (a recap after the training, a mandatory-attendance notice, etc.) and I can adjust.

A Standard Announcement B Mandatory/Compliance C Post-Training Recap

Subject: Upcoming Safety Training - Please Register

Hi all,

We'll be holding a safety training session for the team. Please see the details below and make sure to attend, as this is an important part of maintaining a safe workplace for everyone.

Date: [insert date]
Time: [insert time]
Location: [insert location/link]
Duration: [insert duration]

What to expect:

- Overview of key safety procedures and protocols
- Emergency response guidelines
- Q&A session

Please confirm your attendance by [insert RSVP deadline]. If you have a scheduling conflict, let me know as soon as possible so we can find an alternative time.

AFTER

The same task, with context

What I typed:

*I'm Ms. Example Name from the Lab Safety Office (**who**). Write an email to about 40 lab staff inviting them to a compulsory one-hour chemical safety refresher (**what**) on 14 October at 10:00 in Training Room 2 (**when, where**). They must bring their staff card, and should reply to safety-office@example.org by 9 October if they can't attend. Tone: friendly, but clear that it is required (**tone**). Under 120 words, with a clear subject line (**length**).*

What I got: a finished email, ready to send.

I'm Ms. Example Name from the Lab Safety Office. Write an email to about 40 lab staff inviting them to a compulsory one-hour chemical safety refresher on 14 October at 10:00 in Training Room 2. They must bring their staff card, and should reply to safety-office@example.org by 9 October if they can't attend. Tone: friendly, but clear that it is required. Under 120 words, with a clear subject line.

Subject: Compulsory Chemical Safety Refresher – 14 October, 10:00

Hi everyone,

This is a reminder that all lab staff must attend a one-hour chemical safety refresher on **Tuesday, 14 October at 10:00 in Training Room 2.**

Please bring your **staff card**, as it's required for entry.

Attendance is compulsory. If you have a genuine conflict and cannot attend, please email safety-office@example.org by **9 October** so we can arrange an alternative session.

Thanks for your cooperation — see you there!

Best regards,
Ms. Example Name
Lab Safety Office

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— TODAY

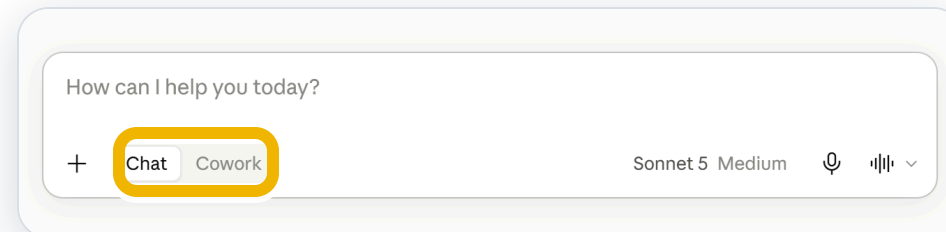
By the end you will be able to ...

- 1 Get a useful answer: give context, say what you want back, ask for changes.
- 2 Let Claude work on a folder of files, while you approve or refuse each change.
- 3 Save a task you repeat as a **skill**, and know when a big job is worth handing to many Claude helpers at once (a **workflow**).
- 4 Know what never goes into Claude, and check what comes out.

— QUICK CHECK

Is everyone signed in?

A · Chat and Cowork



You see **Chat** and **Cowork** (Claude working on a folder of your files; paid plans) in the message box. Most people today.

B · One box

No Chat / Cowork choice in the message box (some Pro and Max accounts).

Nothing to choose: just describe the job. Everything else is the same.

C · No app?

Can't install on your laptop? Open **claude.ai** in your browser. The Chat parts work the same.

Practice files: **Documents** > **Claude-Practice**. Missing? Ask a TA for the USB stick.

Many Claude helpers on one question

This uses a Claude feature called a **workflow** (not the everyday word). Many helpers work on one research question at the same time.

`/deep-research` is a ready-made research job. I start it in the **Code** tab (where Claude works on a folder; more later). You don't need to write code.

It takes a while. We come back to it near the end.

/deep-research What are proven, practical ways to cut energy use in university research labs? Cover fume hoods, freezers, equipment scheduling and lighting, and include sources.

deep-research

Workflow 34 agents 2m 46s

The deep-research workflow is now running in the background. It will:

1. Split your question into 5 search angles: fume hoods, freezers, equipment scheduling, lighting, and lab-wide programs.
2. Run parallel web searches and fetch the top ~15 sources.
3. Pull out specific claims that can be checked, such as "a Sash-closing campaign saves X kWh," and have three independent checkers try to disprove each one. A claim is dropped if two of the three refute it.
4. Write a ranked report with citations.

Safe use first

Five minutes on what never goes in, and the settings that matter.

— THE RULE

Never put these into Claude

Passwords and secret codes

National ID, Iqama or passport numbers

Health information

Student records and grades

Personal data about other people

Contracts and tenders

Anything marked confidential or internal

Unpublished research and patent material

Data under an agreement (NDA or data-use agreement)

Don't put sensitive information into Claude. Follow KAUST policy on what you can and can't use.

— YOUR ACCOUNT

Your account decides the rules

Personal: Free, Pro, Max

Your chats (and your work in Claude Code, which comes later) can be used to improve Claude, unless you switch that off.

Switch on: kept up to 5 years.

Switch off: not used, and a chat you delete is gone within 30 days.

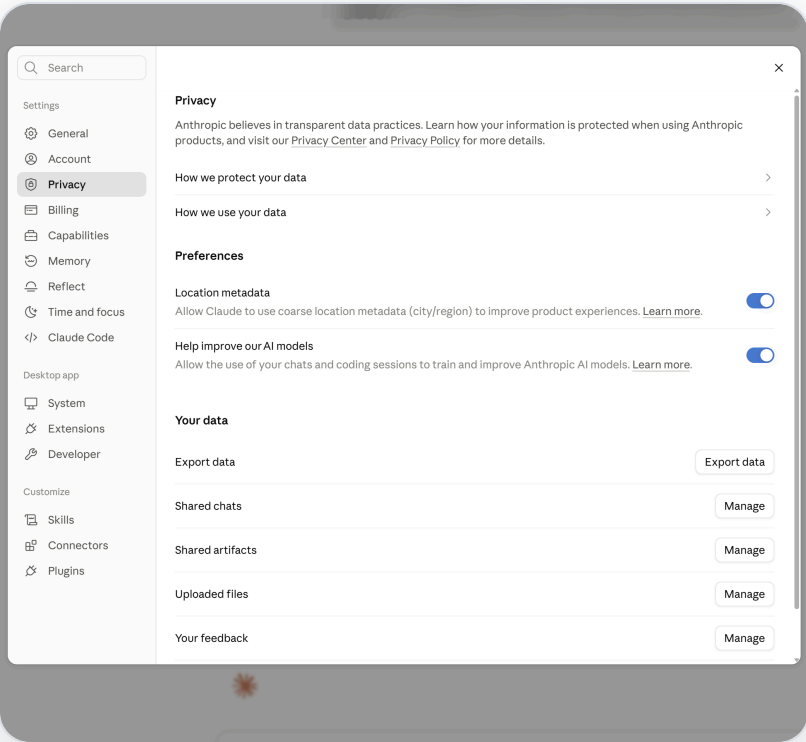
Work: Team, Enterprise, Education

Not used to improve Claude. Your organisation's admin controls the settings, and may be able to download members' chats.

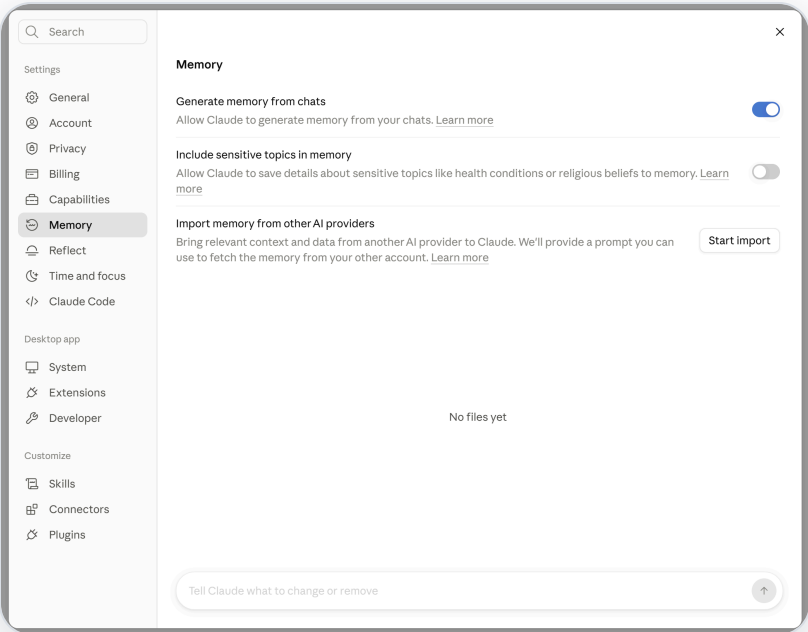
Signing in with your KAUST email does **not** make it a work account. Today, everyone is on a personal account.

YOUR SETTINGS

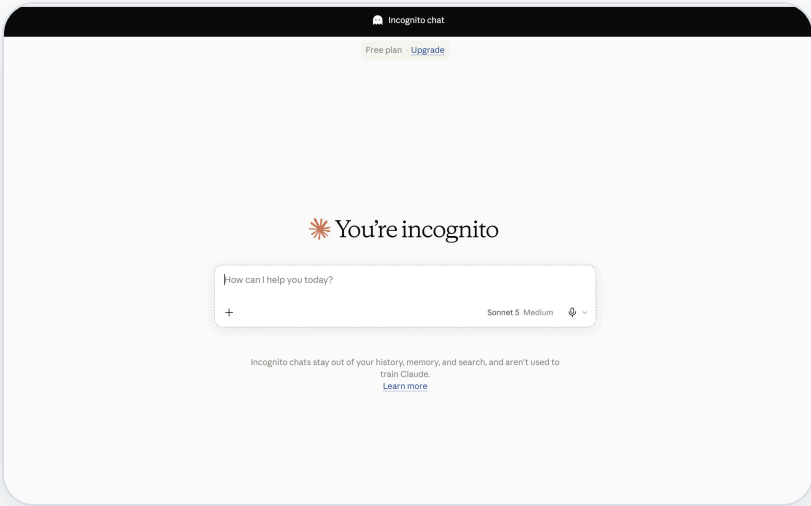
Four privacy screens to know



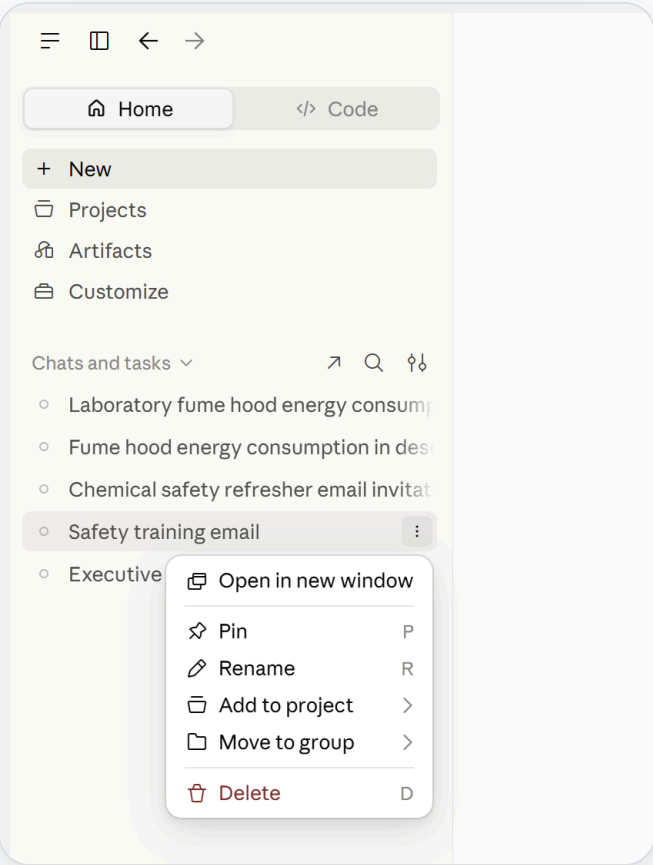
Help improve our AI models: lets Anthropic, the company that makes Claude, use your chats to improve it.



Memory: what Claude remembers about you from past chats.



Incognito (ghost icon): a chat that stays out of your history and memory.



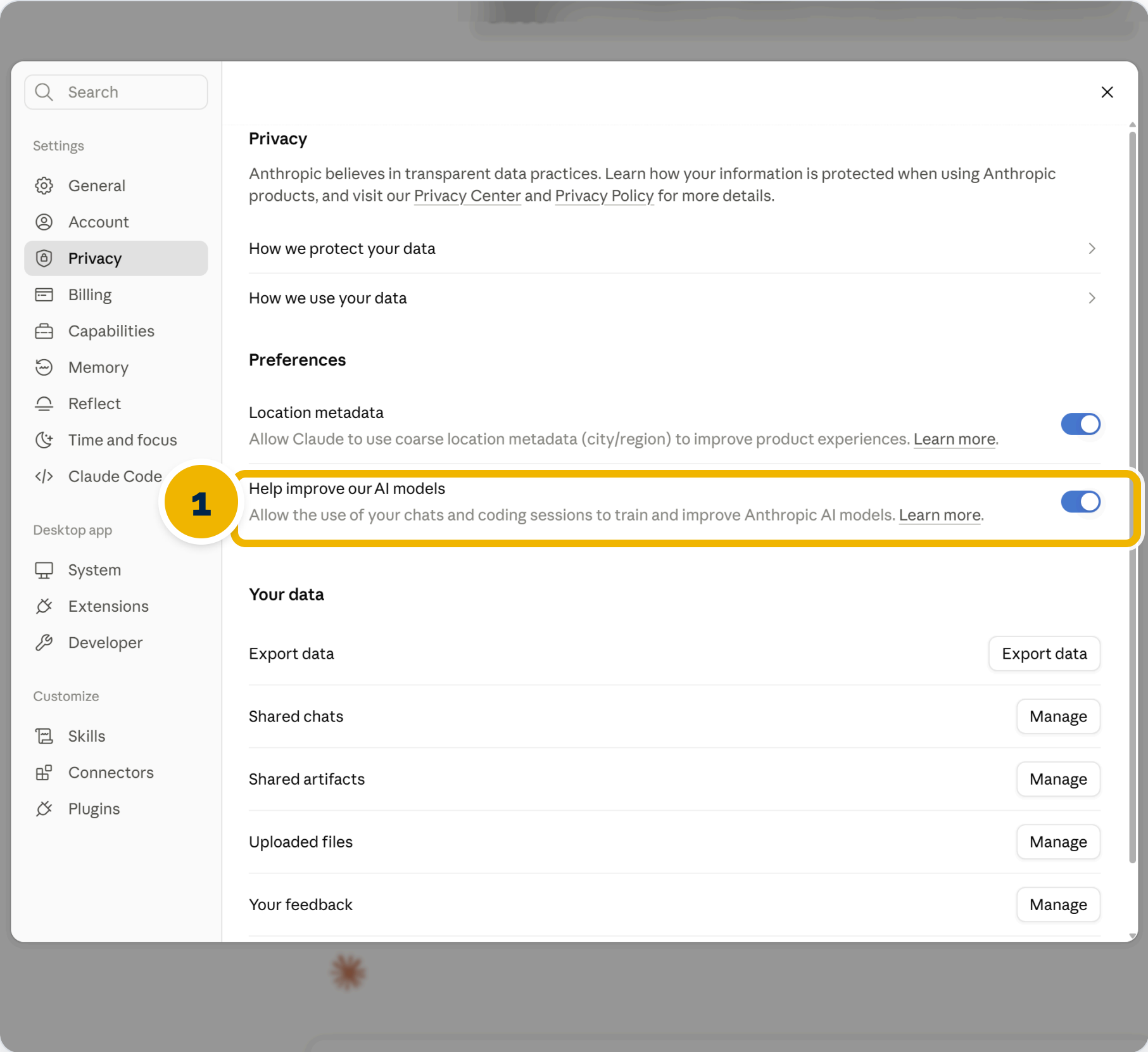
Delete: removes a chat from your list (and from Anthropic within 30 days if the switch is off).

Settings > Privacy

Click your initials (bottom left), then **Settings**, then **Privacy**.

Help improve our AI models: "Allow the use of your chats and coding sessions to train and improve Anthropic AI models."

Off means your chats are not used.

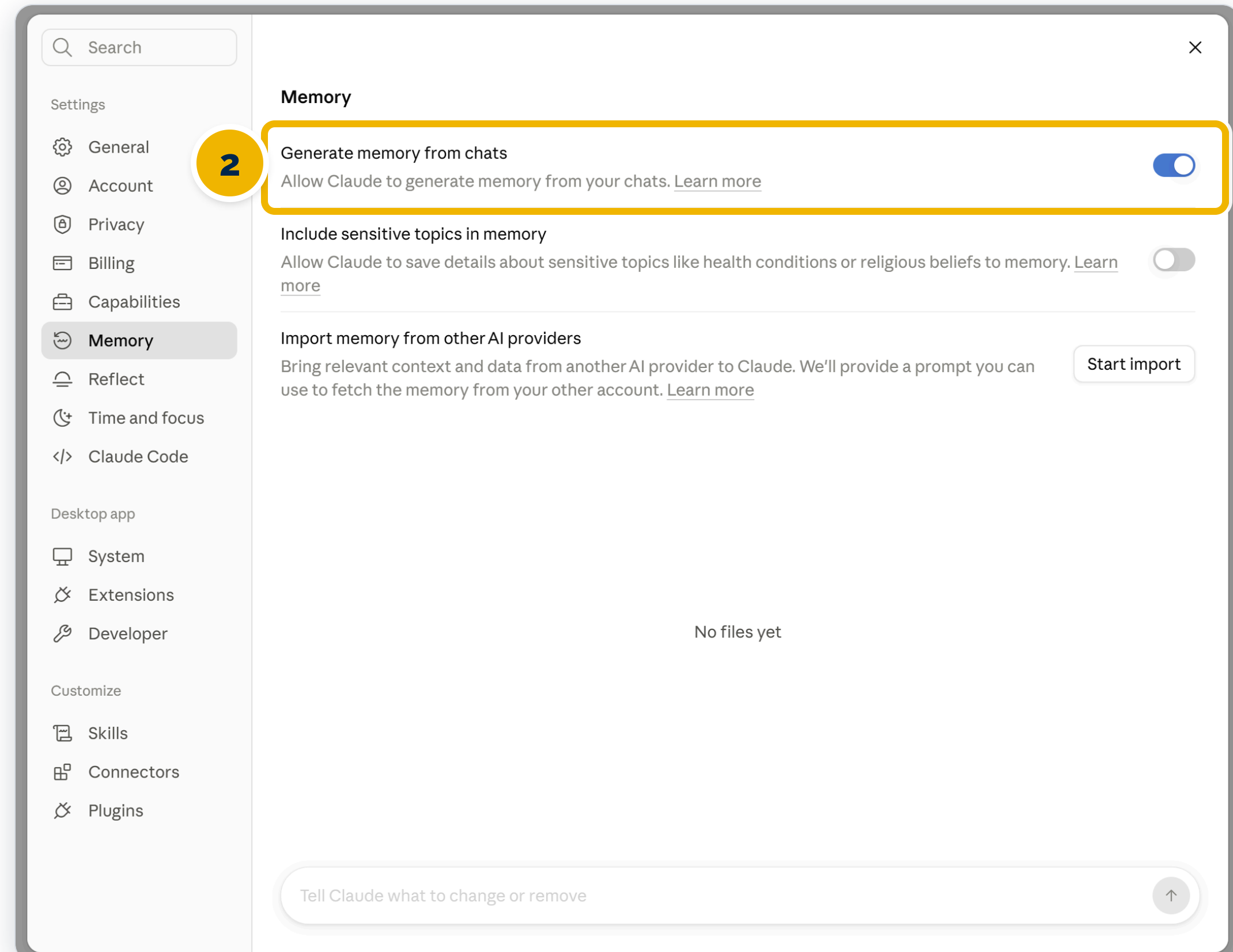


Settings > Memory

Generate memory from chats:

Claude remembers things about you across chats.

Deleting a chat does **not** delete what memory learned from it. Remove it here, or type in *"Tell Claude what to change or remove"*.

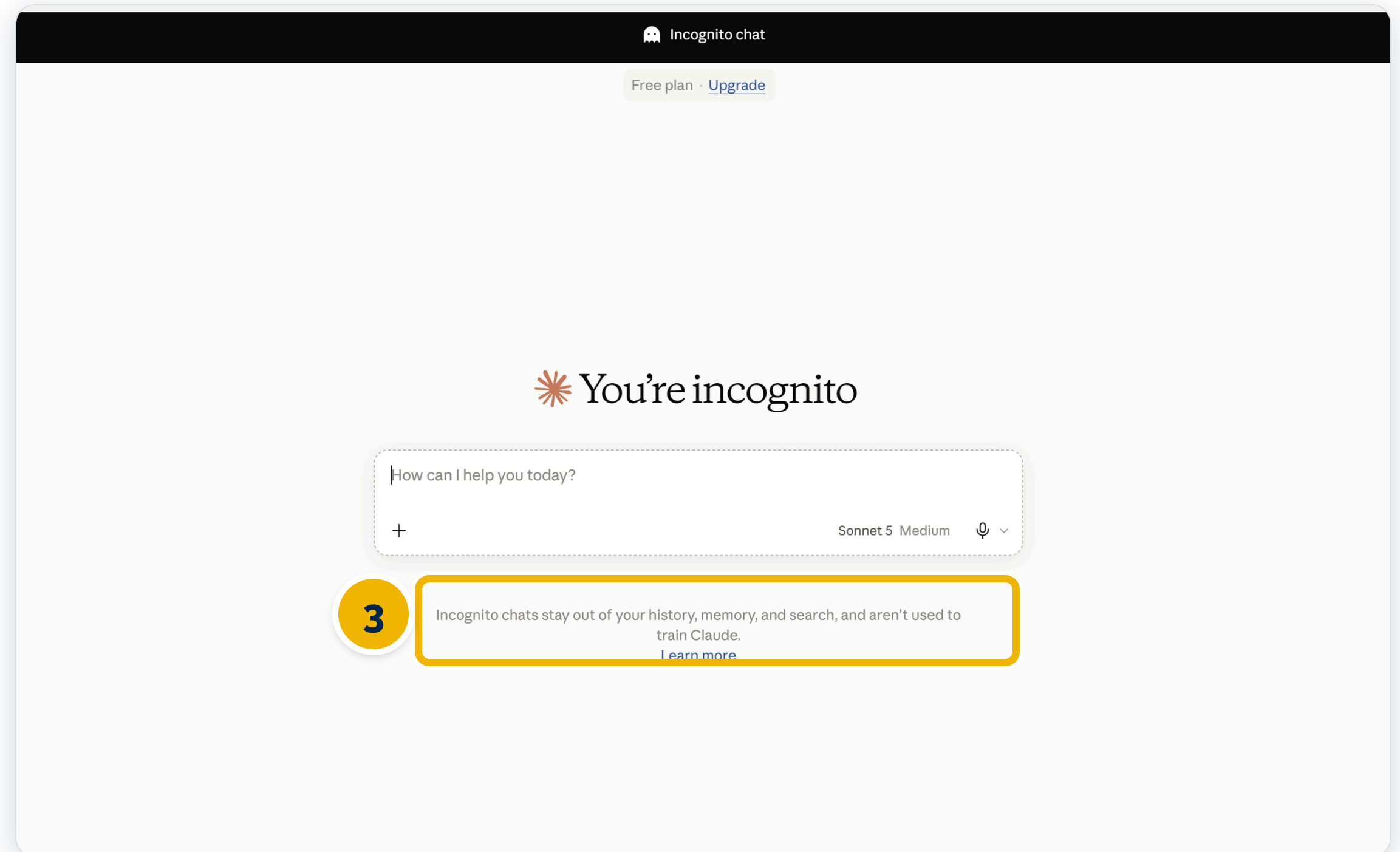


Incognito (the ghost icon)

Click the **ghost icon** at the top right of a new chat.

"Incognito chats stay out of your history, memory, and search, and aren't used to train Claude."

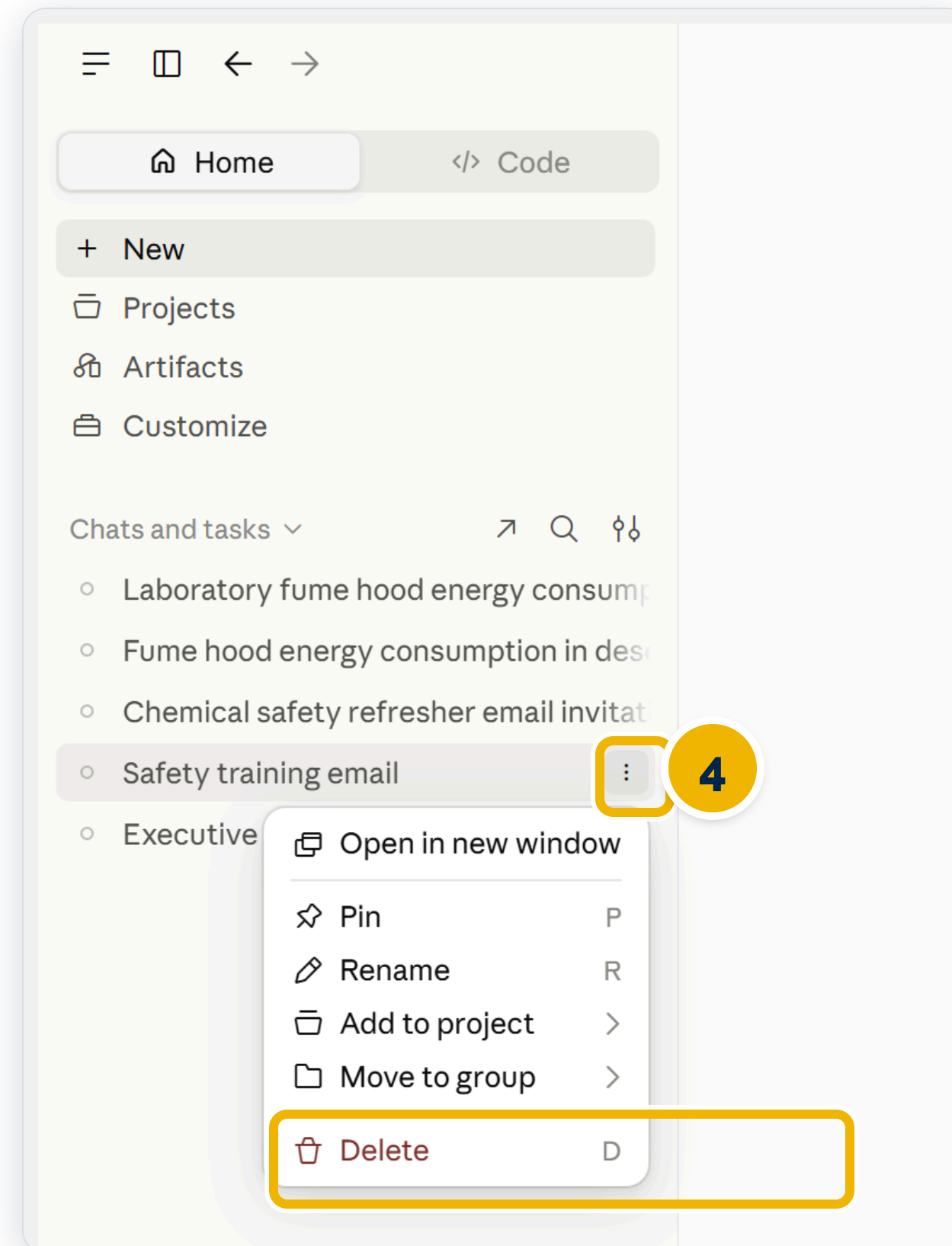
They are still kept for 30 days. The never-put list still applies.



Deleting a chat

Hover a chat in the sidebar, click :, then **Delete**.

It leaves your list straight away. With the improve switch off, Anthropic also deletes it within 30 days.



— TWO SURPRISES

Thumbs up or down sends the whole chat

Give negative feedback

What type of issue do you wish to report? (optional)

Select...

Please provide details: (optional)

What was unsatisfying about this response?

Submitting this report will send the entire current conversation to Anthropic for future improvements to our models. [Learn More](#)

Cancel Submit

Rating a reply sends the **entire conversation** to Anthropic. It can be kept up to 5 years and used to improve Claude, **even with the switch off**. Don't rate a chat that holds anything sensitive.

And incognito is not a safe place for secrets

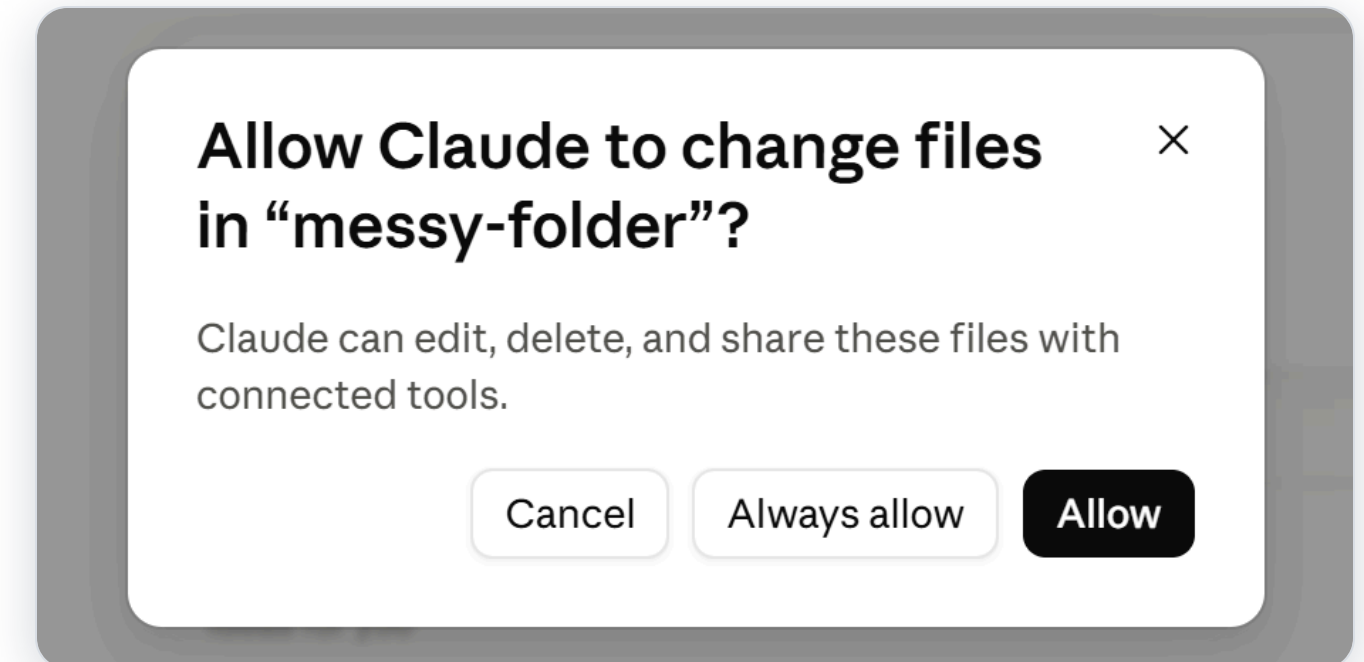
It keeps a chat out of your history and memory, but Anthropic still keeps it for 30 days.

The never-put list still applies.

— YOUR FILES

When Claude can open your files

- Anything Claude reads (a file you attach, or files in the folder you give Cowork or Code) is **sent to Anthropic** to be processed.
- Cowork and Claude Code (ways Claude works on a folder of your files; coming later) can **change and delete** files.
- Give them **one practice folder**. Never your whole home folder, Documents or Desktop.
- Read every **"may I do this?"** before you say yes. Keep a backup.
- If Claude Code asks *"Can Anthropic look at your session transcript?"* (the full record of what you and Claude did), answer **No**.



Cowork's own warning when you give it a folder.

— PLUGS AND LINKS

Connect only what you need. Treat links as public.

Connectors

A **connector** is a plug that lets Claude reach another app, like your email or Google Drive.

Once connected, Claude can read what that app holds. Connect only what you need.

Shared links

If you share a chat, or something Claude made, by public link, anyone who gets the link can open it, even if it's forwarded.

Check what Claude gives you **before** you use it or send it.

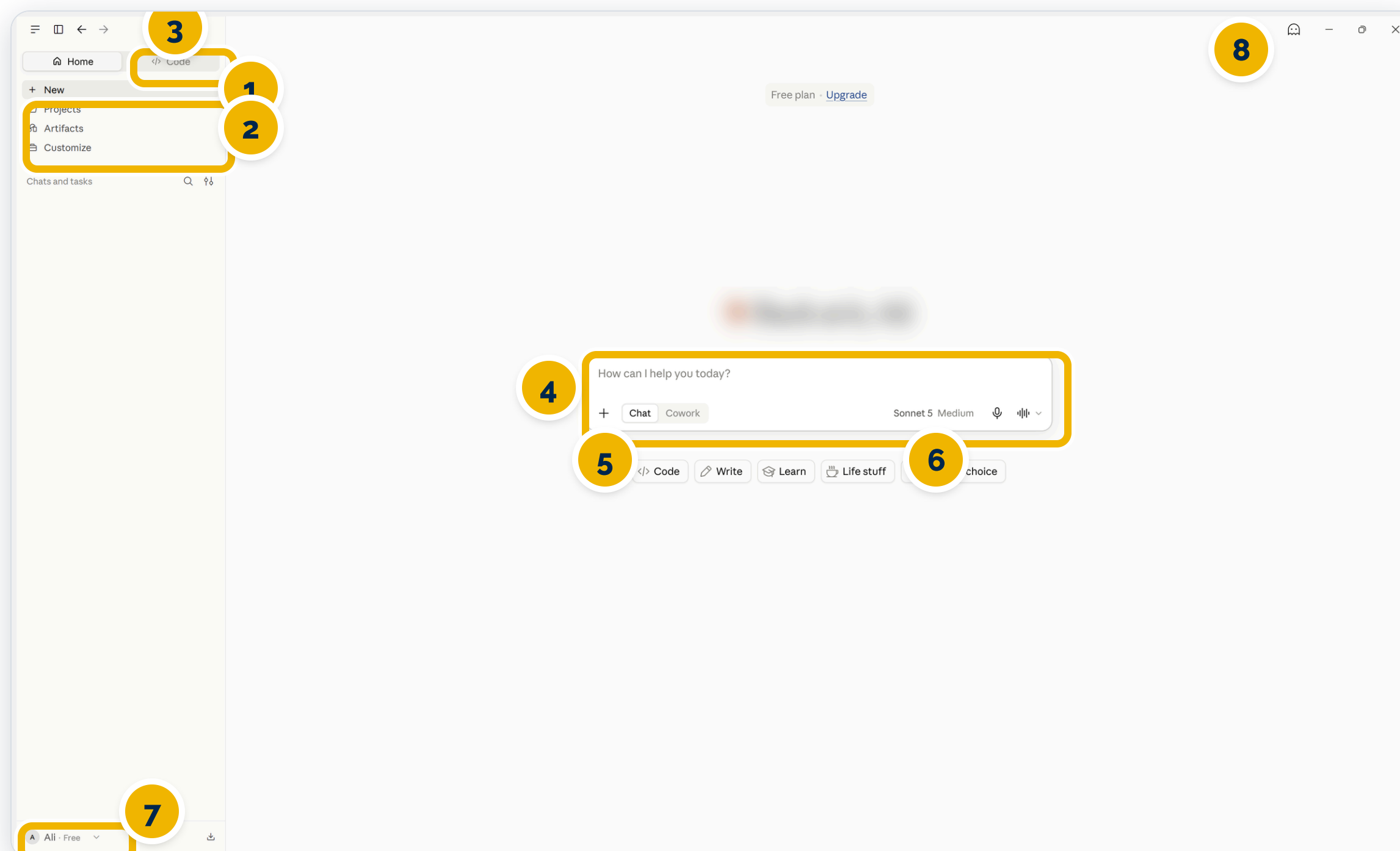
Claude Desktop: Chat

Find your way around, ask well, add files.

THE MAIN SCREEN

Where things are

- 1 **New**: starts a new chat
- 2 Projects, Artifacts, Customize (all explained later)
- 3 The Code tab (Claude works on a folder; later)
- 4 The message box: where you type to Claude
- 5 + adds files, skills, web search
- 6 The model: different versions of Claude, some faster, some more careful
- 7 Your initials: Settings
- 8 Incognito (ghost icon)

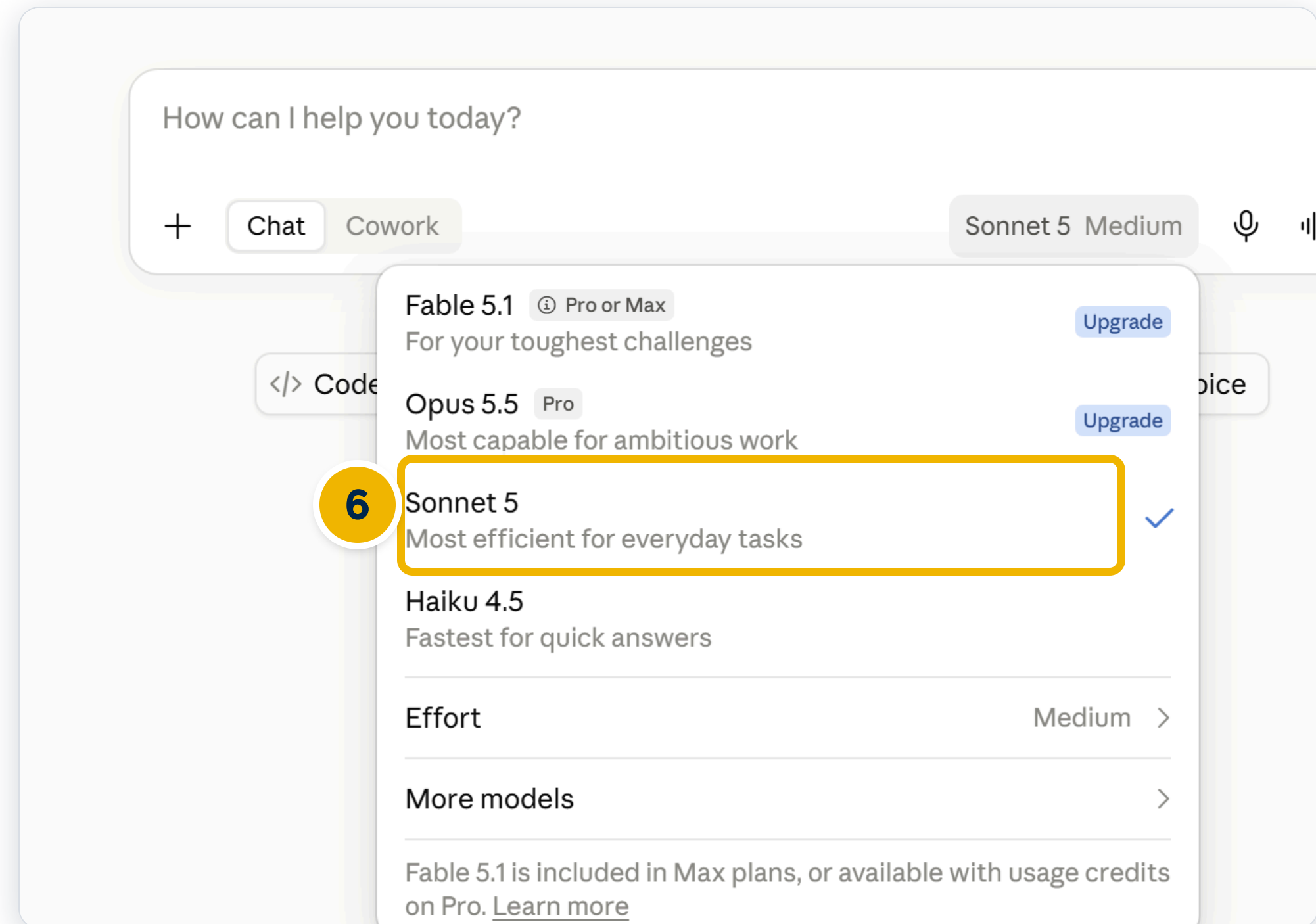


THE MAIN SCREEN · THE MODEL MENU

Different versions of Claude

Click the model name next to the send button.

- **Sonnet 5:** "Most efficient for everyday tasks" (Free)
- **Haiku 4.5:** "Fastest for quick answers" (Free)
- **Opus 5.5:** "Most capable for ambitious work" (Pro and up)
- **Fable 5.1:** "For your toughest challenges" (Max, or Pro with usage credits)



— ASKING WELL

Three habits

1 · Give context

Who is it for? Why do you need it?
What do you already have?

"I work in the admin office of a research institute..."

2 · Say what you want back

Format, length, language, tone.

"5 bullet points, under 150 words."

3 · Ask for changes

It's a conversation. The first answer is a draft.

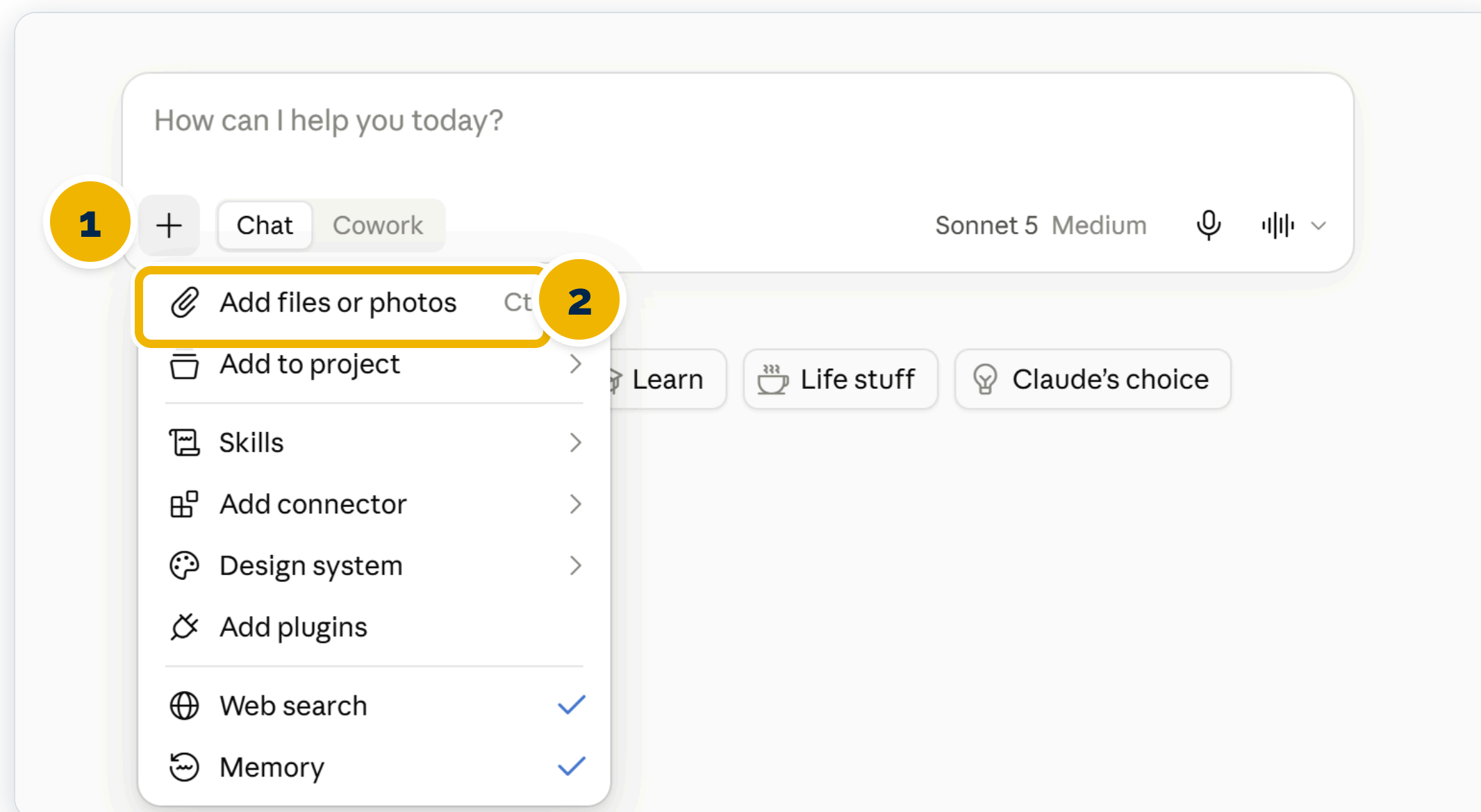
"Make it shorter and put the numbers in a table."

— ADDING FILES

Give Claude the file, not a description of it

Click **+**, then **Add files or photos**. Or drag the file into the chat.

PDF, Word, Excel, images and more.



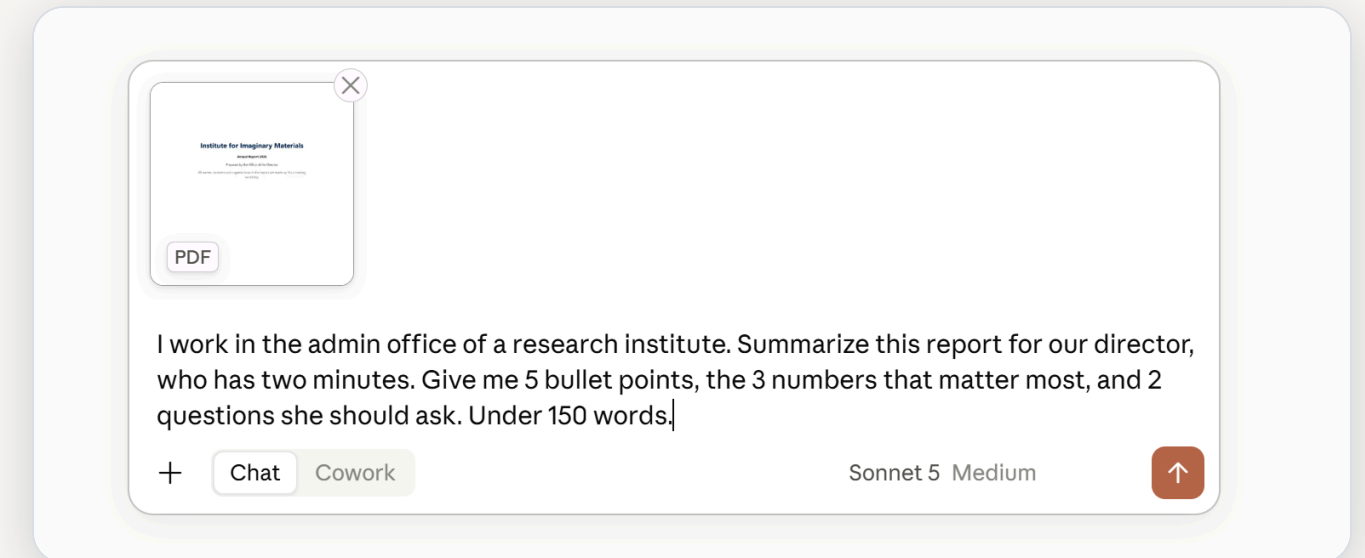
Your turn 1

● 8 minutes

Works on Free

Summarize a report, then ask for a change

- 1 Click **New** (top of the sidebar).
- 2 Click **+**, then **Add files or photos**. Open **01-ask-well** inside Claude-Practice and pick **annual-report-2026**.
- 3 Copy the Step 3 text from **text-to-copy.txt** (same folder), paste it, press Enter.
- 4 Read the answer. Check one number against the table on page 3 of the report.
- 5 Ask for a change: copy the Step 5 text and send it.



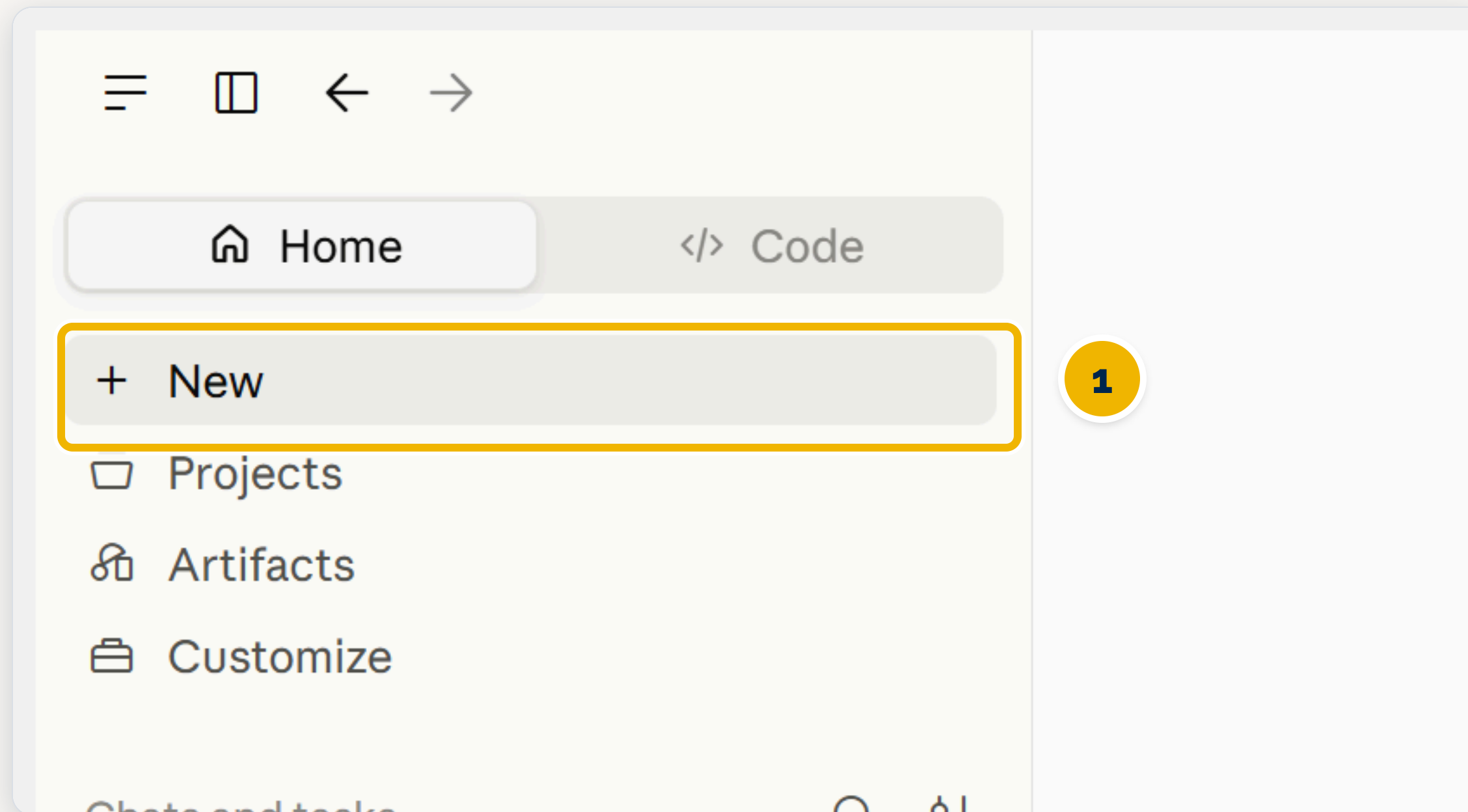
Stuck? Can't sign in to the app? Use claude.ai in your browser, or pair with a neighbour. Can't find the file? Drag it from the folder into the chat. Windows may hide the end of file names (.pdf, .txt). That's fine.

If your screen looks different: Claude may make a file instead of answering in the chat. That's fine: open it.

Your turn 1**STEP 1 OF 5****Works on Free**

Stuck? Go back up to the first "Your turn 1" slide, or ask a TA.

Click New



Your turn 1

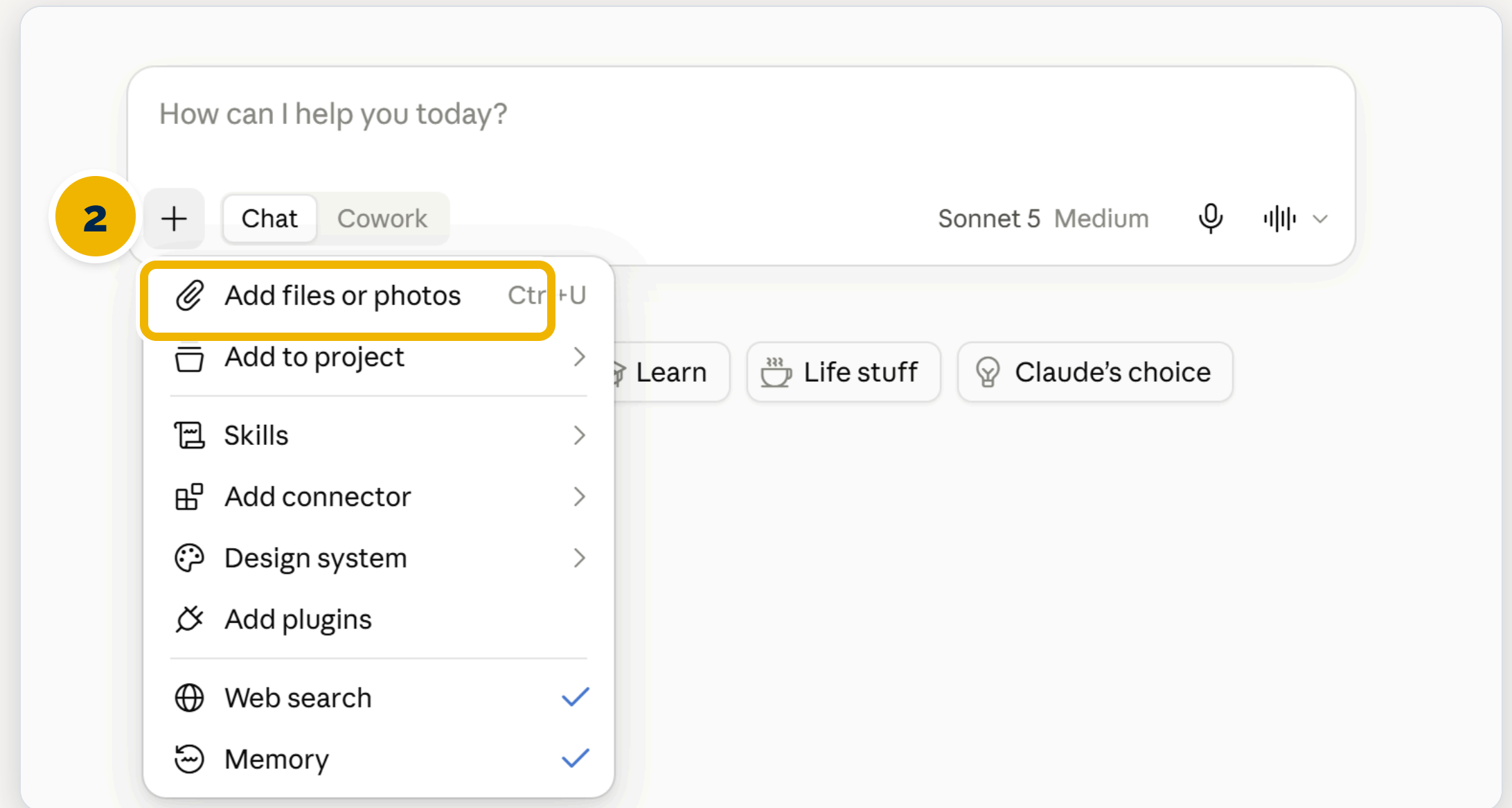
STEP 2 OF 5

Works on Free

Stuck? Go back up to the first "Your turn 1" slide, or ask a TA.

+ then Add files or photos

Open **01-ask-well** inside **Claude-Practice** and pick **annual-report-2026**.



Your turn 1

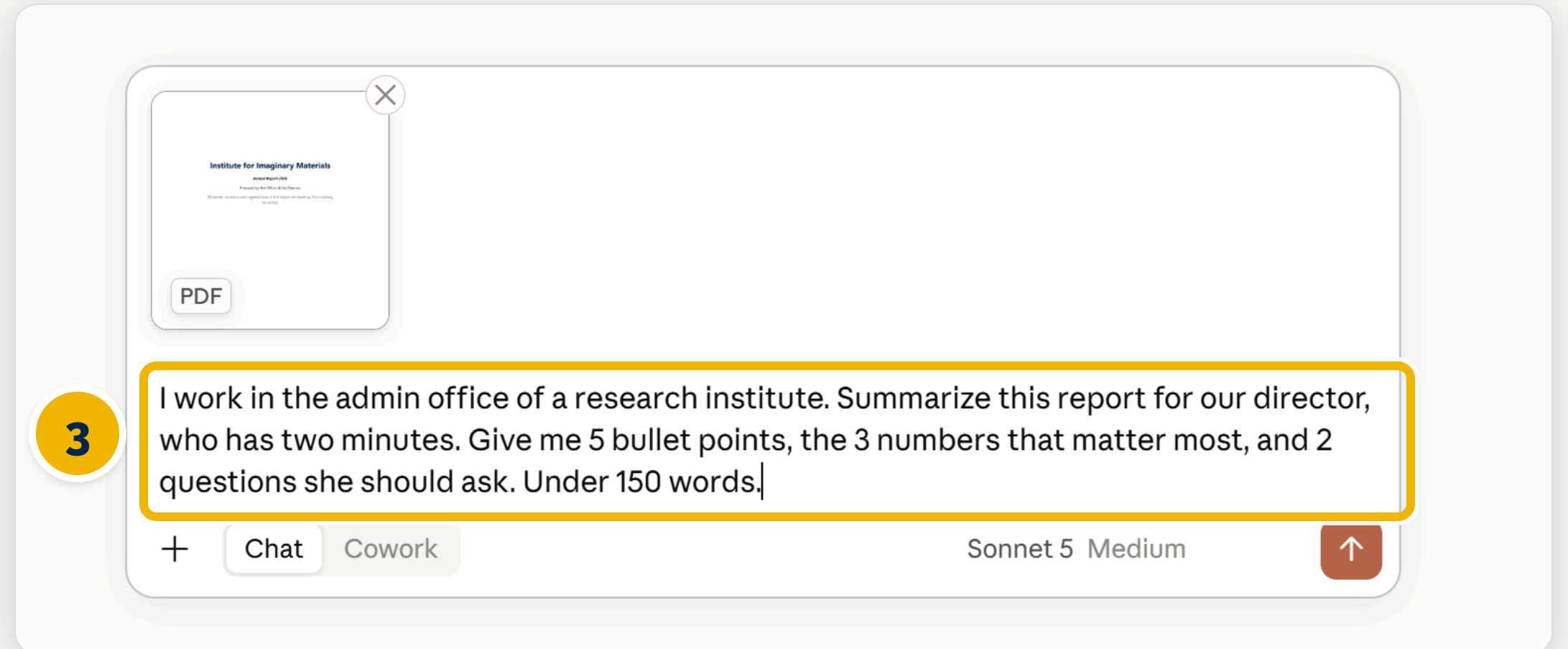
STEP 3 OF 5

Works on Free

Stuck? Go back up to the first "Your turn 1" slide, or ask a TA.

Paste the request and press Enter

I work in the admin office of a research institute. Summarize this report for our director, who has two minutes. Give me 5 bullet points, the 3 numbers that matter most, and 2 questions she should ask. Under 150 words.



Your turn 1

STEP 4 OF 5

Works on Free

Stuck? Go back up to the first "Your turn 1" slide, or ask a TA.

Check one number against the report



I work in the admin office of a research institute. Summarize this report for our director, who has two minutes. Give me 5 bullet points, the 3 numbers that matter most, and 2 questions she should ask. Under 150 words.

5 key points:

- Funding, papers, students, staff, and partners all grew in 2026 — a strong year overall.
- Lab energy use dropped 12%, hitting the sustainability target early.
- Lab 3's move ran four months late due to a ventilation equipment delay, pausing three projects.
- Technical staff turnover stayed high at 14%.
- 2027 goals: cut energy 8% more, turnover below 10%, launch the water-treatment pilot, reach 30 industry partners.

3 numbers that matter most:

- SAR 48.2M in research funding (+16%)
- 38% of funding concentrated in just two grants
- 14% technical staff turnover

4

2 questions to ask:

1. What's the plan to diversify funding beyond the two dominant grants?
2. What specifically will fix the single-supplier risk that caused the Lab 3 delay?



annual-report-2026, page 3: The year in numbers

Measure	2026	2025	Change
Research funding received	SAR 48.2 million	SAR 41.5 million	+16%
Peer-reviewed papers	214	198	+8%
Graduate students	312	287	+9%
Staff (full-time)	145	139	+4%
Industry partners	23	17	+35%
Lab energy use	7.9 GWh	9.0 GWh	-12%
Public events	41	33	+24%
Visitors to open days	6,800	5,100	+33%

Page 3 of annual-report-2026 (made-up data).

Your turn 1

STEP 5 OF 5

Works on Free

Stuck? Go back up to the first "Your turn 1" slide, or ask a TA.

Ask for a change

Put the 3 numbers in a table, with one line on what each means. Then make the rest shorter: 80 words at most.

5

Put the 3 numbers in a table, with one line on what each means. Then make the rest shorter: 80 words at most.

Number	What it means
SAR 48.2M	Research funding, up 16% from 2025
38%	Share of funding from just two grants
14%	Technical staff turnover this year

Highlights: Strong growth across funding, papers, students, and partners; energy use down 12% (target met early); Lab 3 move ran 4 months late (supplier issue); 2027 goals include lower turnover and the water-treatment pilot.

Ask her:

1. Plan to diversify beyond two dominant grants?
2. Fix for the single-supplier ventilation risk?

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CHECKING ANSWERS

It can sound sure and still be wrong

I asked, with web search off:

From memory, without searching: give me five frequently cited peer-reviewed papers about the energy use of laboratory fume hoods, with authors, year, journal and DOI.

From memory, without searching: give me five frequently cited peer-reviewed papers about the energy use of laboratory fume hoods, with authors, year, journal and DOI.

Recalling fume hood energy studies while flagging citation uncertainty. >

A few caveats up front: I don't have search access in this mode, and citation details (especially DOIs) are exactly the kind of thing I can get wrong or partially fabricate from memory — please verify each of these before relying on them.

With that said, here's my best recollection of frequently cited work on lab fume hood energy use:

- Mills, E., & Sartor, D. (2005).** "Energy use and savings potential for laboratory fume hoods." *Energy*, 30(10), 1859–1873. DOI: 10.1016/j.energy.2004.11.008

The official record (Crossref, which keeps the DOI of every paper: its permanent ID number):

Metadata Search
Funding Data
Link References
Status
API
Help
Sign in

SORT BY:
RELEVANCE
PUBLICATION YEAR

PAGE 1 OF 1 RESULTS

Showing DOI matching 10.1016/j.energy.2004.11.008

Energy use and savings potential for laboratory fume hoods

JOURNAL ARTICLE published July 2005 in *Energy*

Authors: Evan Mills | Dale Sartor

	Claude said	The record
Title, authors, year, journal	✓	✓
Pages	1859–1873	1859–1864

Five checks before you rely on it

- 1 Numbers: compare with the file you gave it.
- 2 References: search the exact title in the KAUST Library or Google Scholar, or paste the DOI (the paper's ID number) into doi.org.
- 3 Quotes and names: find them in the original source.
- 4 Ask Claude: *"Which parts are you unsure about?"* and *"Quote the sentence you used."*
- 5 For sources, turn on web search: answers come with links you can open.

Under every chat Claude itself says: "Claude is AI and can make mistakes. Please double-check responses."

CHECKING ANSWERS · A GOOD SIGN

With web search, it said "I can't find that"

Same kind of question, with web search on:
Claude searched, found no such studies, and said so.

"I don't want to hand you fabricated citations with invented DOIs just to fill three slots."

Still: check what it does give you.

Give me three peer-reviewed studies, with authors, year, journal and DOI, that measured how much energy fume hoods use in university labs in hot desert climates.

Searched the web >

I searched, but I can't find three peer-reviewed studies that specifically measured fume hood energy use in university labs located in hot desert climates. Here's what's actually out there:

What exists:

- Evan Mills & Dale Sartor's foundational study, "Energy Use and Savings Potential for Laboratory Fume Hoods" (*Energy*, 2005), models fume hood energy costs across various global climate zones (LBNL's version is at <https://eta-publications.lbl.gov/sites/default/files/fumehoodssartormills2006.pdf>), but it isn't desert-specific — it compares multiple locations rather than isolating hot arid climates.
- A ScienceDirect study on "Sustainable laboratory evaluations: Optimized fume-hood-intensive ventilation..." (~2021/2022, *Journal of Cleaner Production*) explicitly frames itself around "hot and humid climates" — but that's humid, not arid/desert, and the setting is Singapore.
- Vendor pages, trade press (C&EN, Lab Manager), and commercial fume-hood suppliers in Saudi Arabia/UAE/Qatar all mention high ambient heat as a driver of HVAC load, but none of these are peer-reviewed measurement studies — they're marketing or trade journalism.

What this means: I don't want to hand you fabricated citations with invented DOIs just to fill three slots — that would be actively harmful if you're using this for research or a paper. The intersection of "fume hood energy measurement" + "university lab" + "hot desert climate specifically (not humid)" appears to be a genuine research gap, or at least not something indexed clearly under those terms.

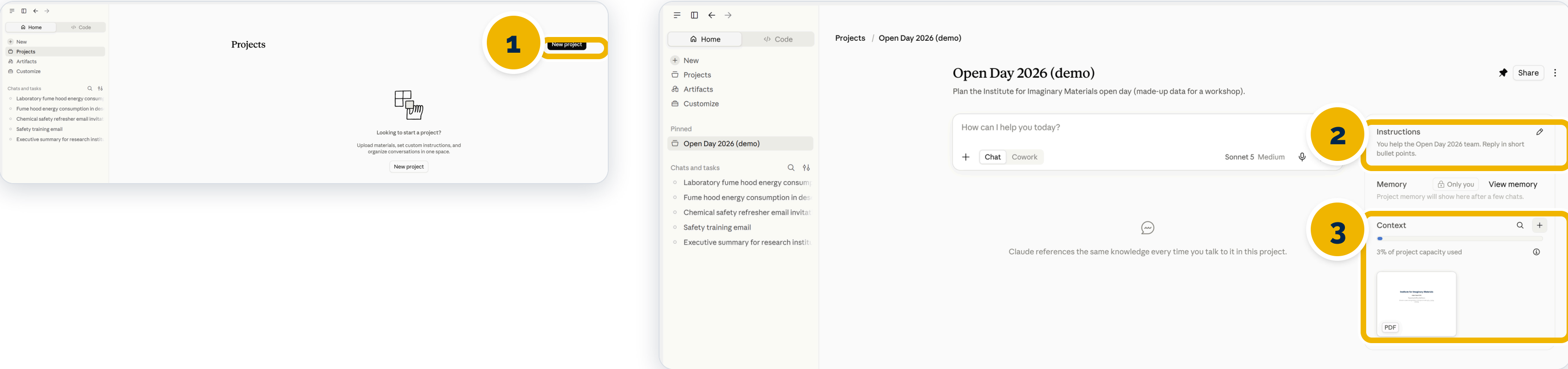
A few ways I could help further:

- Search academic databases more specifically (e.g., via Google Scholar-style queries for institutions like KAUST, King Saud University, Arizona State, UAE University) for building-energy-audit papers that include fume hoods as one HVAC load component, even if not the sole focus
- Broaden to "laboratory HVAC energy" in arid/desert climates generally, which has more literature, and flag where fume hoods are specifically broken out
- Point you to the Mills & Sartor climate-comparison data, which includes desert-adjacent locations (e.g., Phoenix, AZ) in its regional analysis, if a data point rather than a dedicated study would help

PROJECTS

A space for one piece of work

A **Project** keeps the files and instructions for one piece of work, so you don't repeat them in every chat. Free: up to 5 projects.



1 New project · 2 Instructions (how Claude should answer) · 3 Context: files Claude reads in every chat

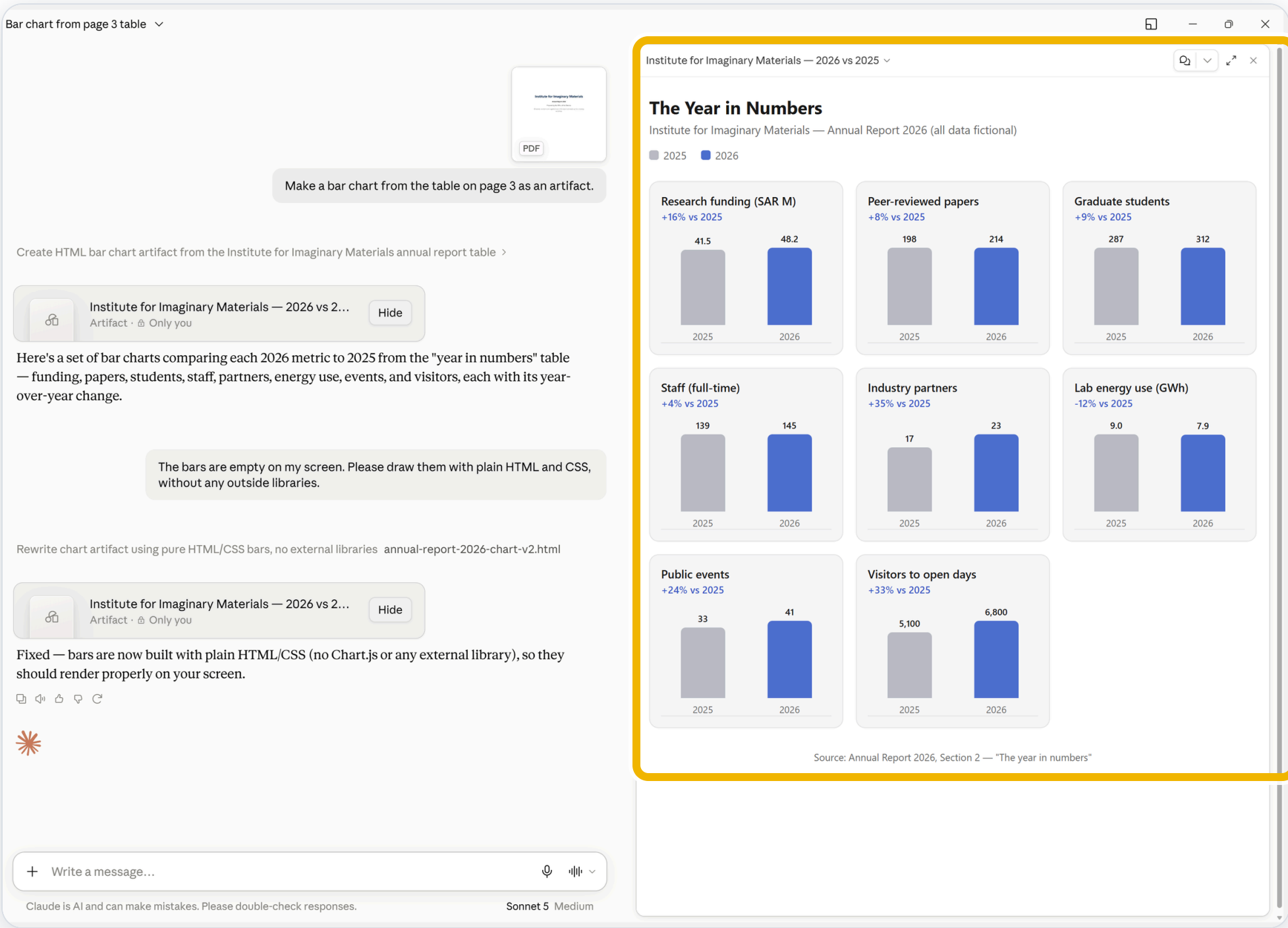
ARTIFACTS

A document, page or chart next to the chat

An **artifact** is a document, page or chart Claude makes that opens next to the chat.

Make a bar chart from the table on page 3 as an artifact.

The first try showed empty bars. I asked for a fix, and got this.



CONNECTORS

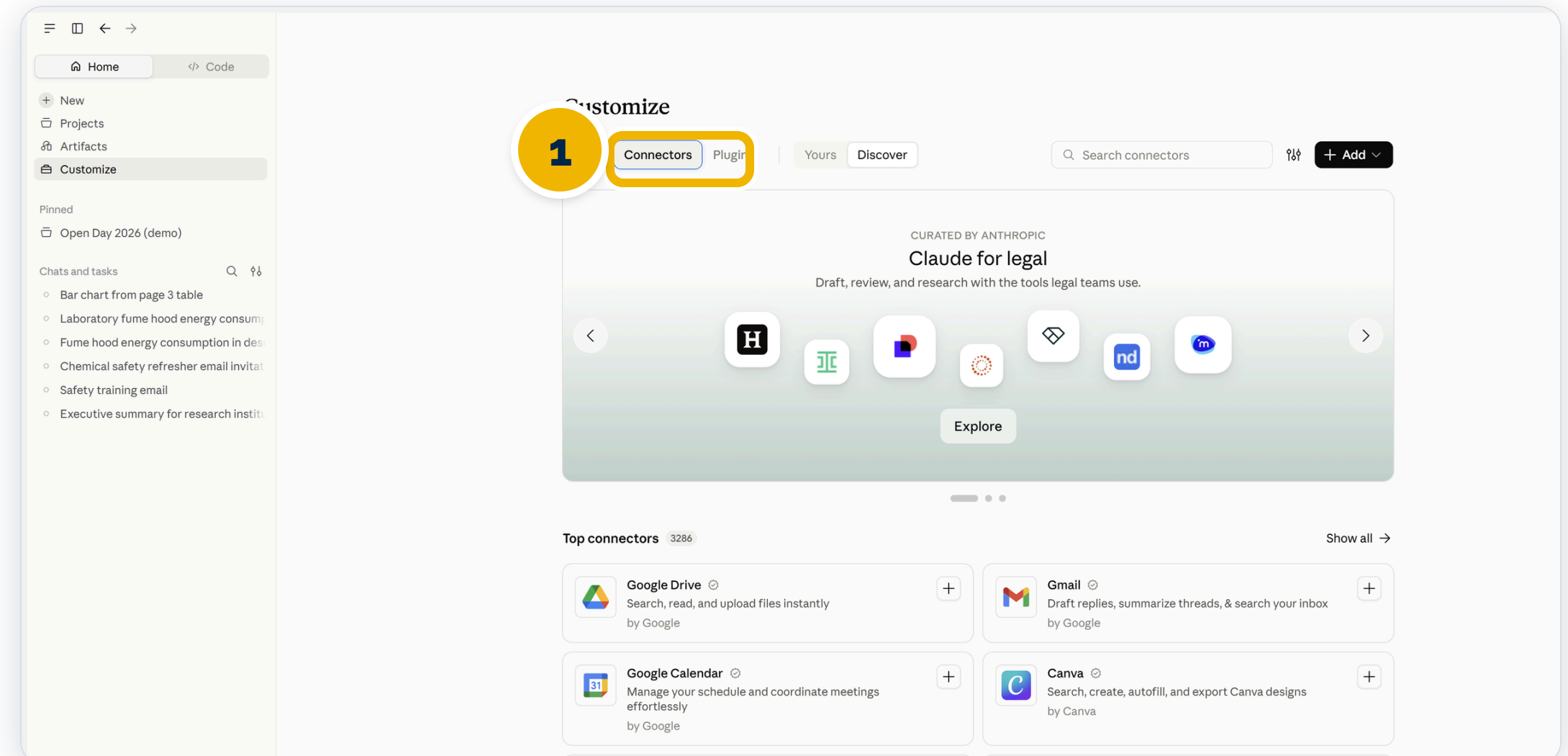
Plugs to your other apps

A **connector** is a plug that lets Claude reach another app, like your email or Google Drive.

Where: **Customize > Connectors**.

Once connected, Claude can read whatever that account can read, for example your whole inbox. Connect only what you need.

Work accounts: an admin may have to allow a connector first. Check KAUST's policy before connecting a KAUST account.



SETTINGS

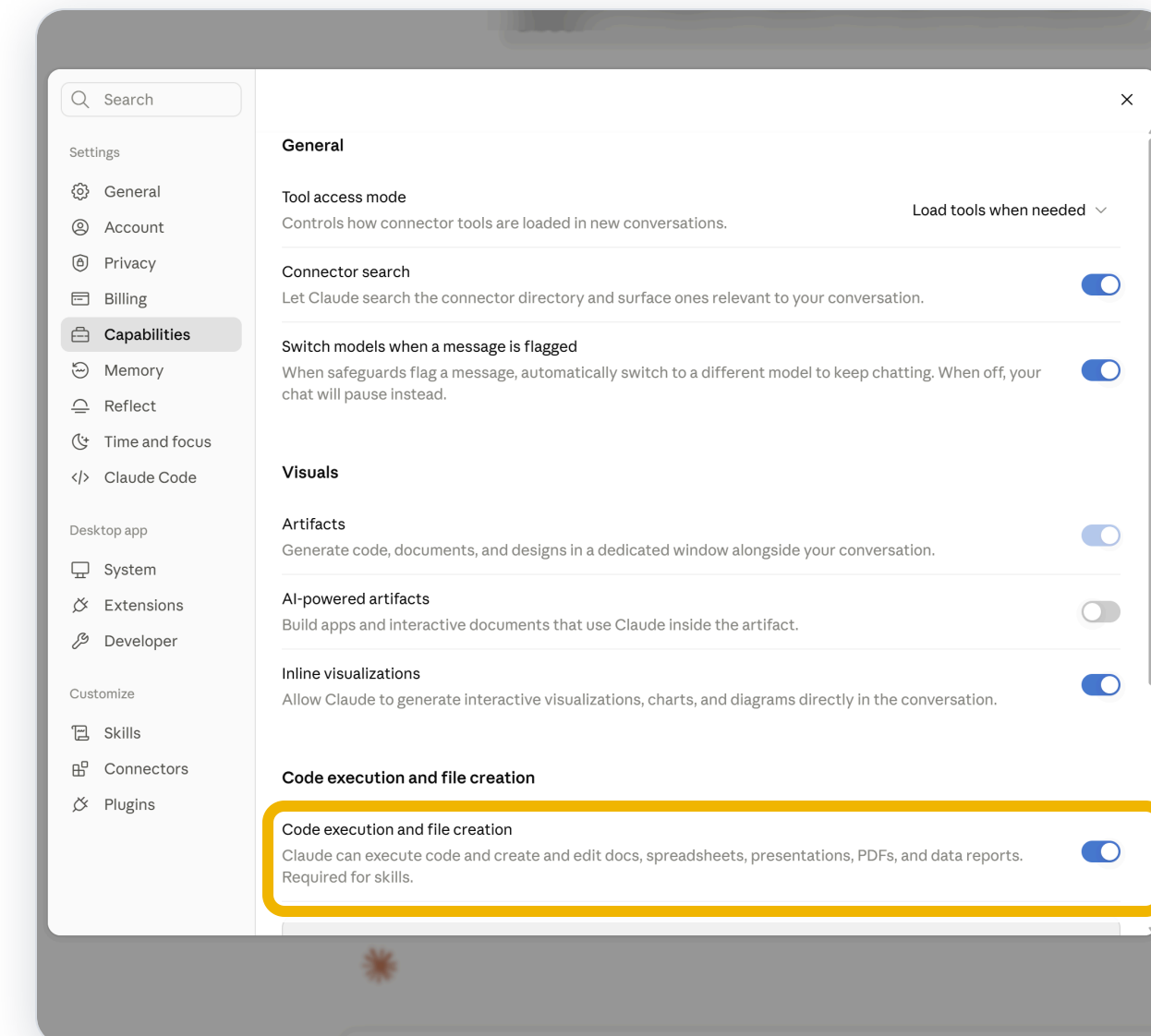
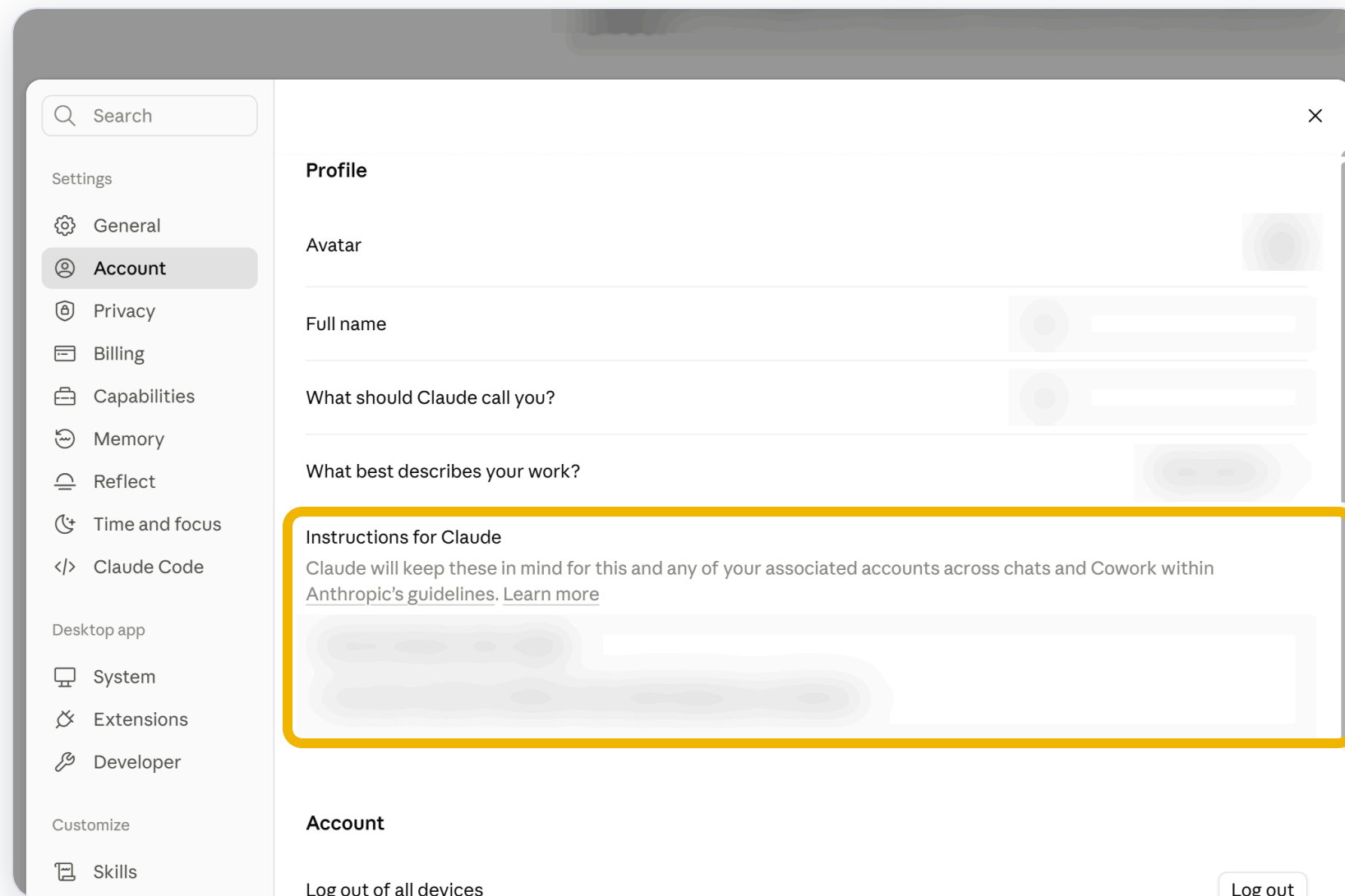
Two settings worth setting once

Instructions for Claude

Settings > Account: tell Claude your role and preferred tone once. It keeps them in mind in every chat.

Code execution and file creation

Settings > Capabilities: lets Claude make real files for you, like Word, Excel or PDF. Skills need it. You won't write any code.



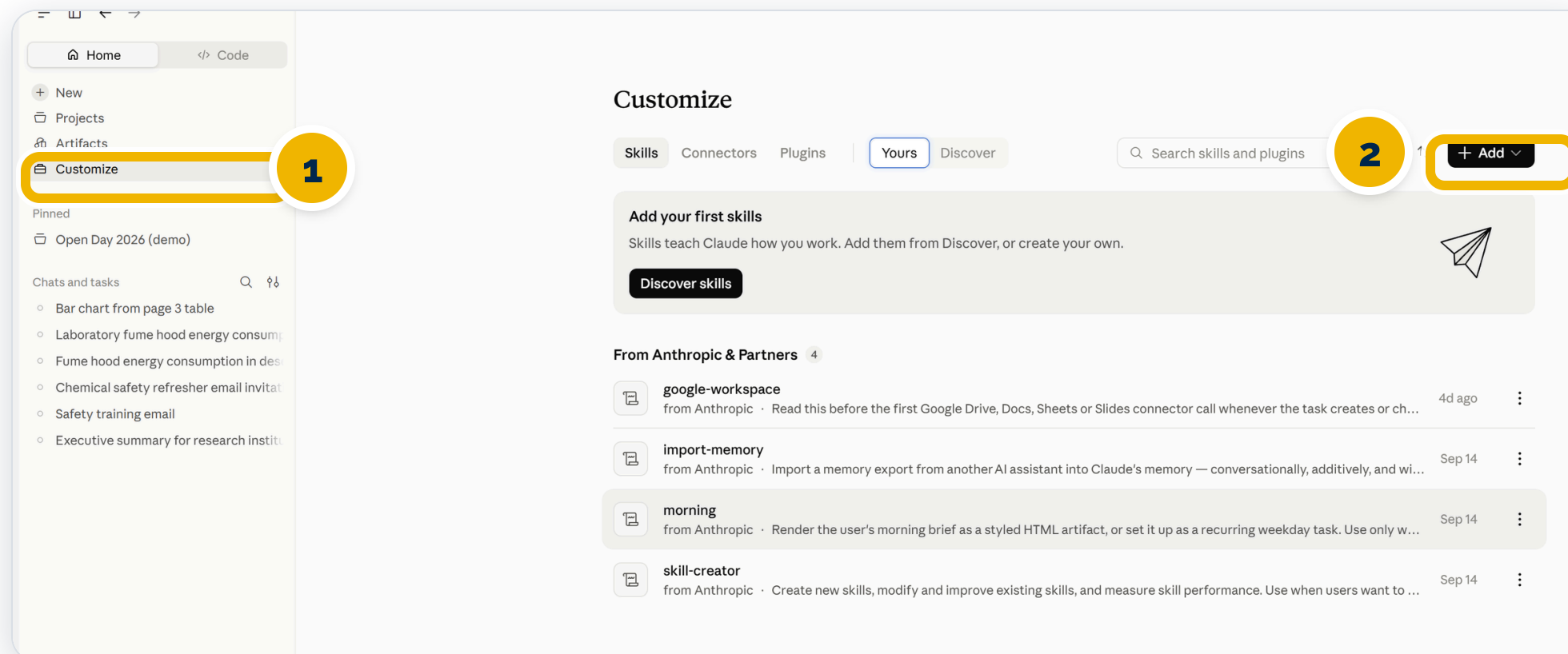
Skills: teach Claude a task once

A saved set of instructions for a task you repeat, so you don't explain it every time.

WHAT A SKILL IS

A saved set of instructions for a task you repeat

Where: **Customize** > **Skills**. Claude uses a skill when your request matches it.



Anthropic's ready-made skills for Word, Excel, PowerPoint and PDF work automatically.

Two skills I use every week

Receipts to expense form: turns a folder of receipt photos into a filled-in expense form.

Proposal drafts: writes a first draft of a KAUST Academy training proposal as a Word file, from the request and a list of courses.

Made once. Now each takes one line to start.

TURNING ON A READY-MADE SKILL

Customize > Skills > Discover > Add

Switch to **Discover**, pick a skill, click **+** (or **Add**). It arrives switched on.

Customize

Skills

Connectors

Plugins

Yours

Discover

🔍

↕

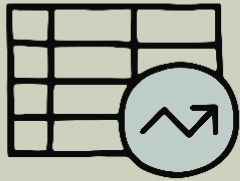
+ Add

From Anthropic

Data

Query, chart and explain your data — SQL, spreadsheets and dashboards in one place.

Add



New skills

Show all →

</>

Render

Deploy, debug, and monitor applications on Render: skills for blueprints, deploys and troubleshooting, a Render...

by Render

+

📈

Amazon Selling Partner

Connect your Amazon Seller Central account to Claude. Ask about sales, inventory, and listing performance an...

by Amazon

+

📢

Pendo Orchestrate

Create, configure and edit draft Pendo Orchestrate email journeys, including multi-email and conditional...

by Pendo

+

📖

learn

Use this skill when the user wants intellectual understanding — learning how or why something work...

by Anthropic

+

📄

doc-coauthoring

Guide users through a structured workflow for co-authoring documentation. Use when user wants to wri...

by Anthropic

+

📄

web-artifacts-builder

Suite of tools for creating elaborate, multi-component claude.ai HTML artifacts using modern frontend web...

by Anthropic

+

Most installed skills

Show all →

Add a ready-made skill and use it

- 1 Check **Settings > Capabilities > Code execution and file creation** is on.
- 2 Click **Customize**, then **Skills**.
- 3 Click **+ Add**, then **Upload skill**.
- 4 Click the box, open **02-skill**, pick **meeting-action-items.zip** (one packed file, zipper icon) as it is, click **Upload**.
- 5 Check the skill is switched on.
- 6 Click **New**, then **+ > Add files or photos** and pick **meeting-notes-open-day** in 02-skill. Type *"Turn these notes into action items."* and press Enter.
- 7 Look for the grey line that mentions **SKILL.md** (the skill's instructions), then the table and "Questions to check".

meeting-notes-open-day.txt
14 lines
TXT

Turn these notes into action items.

Check skill for formatting action items table SKILL.md

#	Owner	Action	Due date	Status
1	Mr. Test Participant	Get 3 quotes from print shops for posters and banners	Thursday 9 Oct	Open
2	Ms. Example Name	Confirm main hall booking in writing	12 Oct	Open
3	Dr. Sample Person	Email department heads asking for 20 volunteers	15 Oct	Open
4	Ms. Example Name	Send budget sheet to finance	"end of next week" (17 Oct?)	Open
5	Mr. Demo Helper	Build open day web page	Not set	Done
6	Unassigned	Check dietary needs of speakers and volunteers	Not set	Open
7	Facilities	Confirm where parking and direction signs can go	Not set	Blocked

Questions to check:

- Budget sheet due date is unclear — noted as "end of next week" (~17 Oct) but Mr. Test thought the 15th.
- Catering/dietary check has no owner — needs to be assigned at the next meeting.
- Parking/signage is blocked on Facilities, with no date for when they'll respond.
- If the main hall booking falls through, the fallback (atrium) has no named owner or deadline.

Stuck? Upload refused: you picked something inside the zip; pick the zip itself (zipper icon). A table but no "Questions to check": type *Use my meeting-action-items skill on these notes*. No Skills page: paste the text from **if-skills-dont-work.txt** instead.

*If your screen looks different: the button may say **Add** or **+**.*

Your turn 2

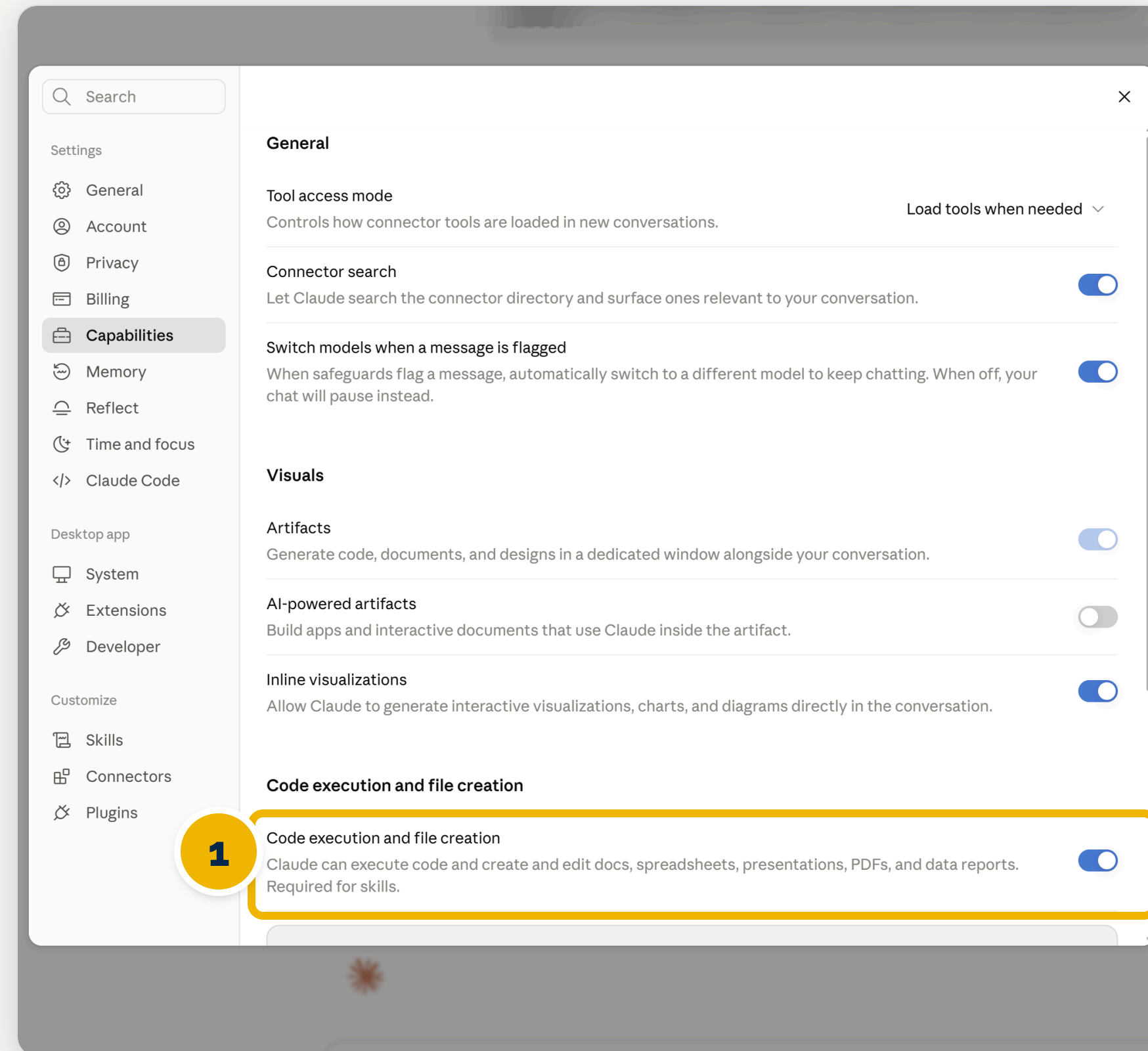
STEP 1 OF 7

Works on Free

Stuck? Go back up to the first "Your turn 2" slide, or ask a TA.

Check the setting skills need

Your initials > **Settings** > **Capabilities**.
Code execution and file creation
should be on. It is on by default.



Your turn 2

STEP 2 OF 7

Works on Free

Stuck? Go back up to the first "Your turn 2" slide, or ask a TA.

Customize, then Skills

Home

Code

New

Projects

Artifacts

Customize

Pinned

Open Day 2026 (demo)

Chats and tasks

Bar chart from page 3 table

Laboratory fume hood energy consump

Fume hood energy consumption in des

Chemical safety refresher email invit

Safety training email

Executive summary for research institu

2

Customize

Skills

Connectors

Plugins

Yours

Discover

Search skills and plugins

+

Add

▼

Add your first skills

Skills teach Claude how you work. Add them from Discover, or create your own.

Discover skills

From Anthropic & Partners 4

google-workspace

from Anthropic · Read this before the first Google Drive, Docs, Sheets or Slides connector call whenever the task creates or ch...

4d ago

⋮

import-memory

from Anthropic · Import a memory export from another AI assistant into Claude's memory — conversationally, additively, and wi...

Sep 14

⋮

morning

from Anthropic · Render the user's morning brief as a styled HTML artifact, or set it up as a recurring weekday task. Use only w...

Sep 14

⋮

skill-creator

from Anthropic · Create new skills, modify and improve existing skills, and measure skill performance. Use when users want to ...

Sep 14

⋮

Your turn 2

STEP 3 OF 7

Works on Free

Stuck? Go back up to the first "Your turn 2" slide, or ask a TA.

+ Add, then Upload skill

The screenshot shows the Claude web interface. On the left is a sidebar with navigation options: Home, Code, New, Projects, Artifacts, and Customize (which is selected). Below these are 'Pinned' items (e.g., 'Open Day 2026 (demo)') and 'Chats and tasks' (e.g., 'Bar chart from page 3 table'). The main area is titled 'Customize' and has tabs for Skills, Connectors, Plugins, Yours, and Discover. A search bar for 'Search skills and plugins' is present. A yellow circle with the number '3' is placed over the '+ Add' button. A yellow box highlights the dropdown menu that appears when the button is clicked, showing three options: 'Upload skill', 'Create a skill', and 'Create with Claude'. Below the 'Add' button, there is a section 'Add your first skills' with a 'Discover skills' button. Further down, a list of skills from Anthropic & Partners is shown, including 'google-workspace', 'import-memory', 'morning', and 'skill-creator'.

Your turn 2

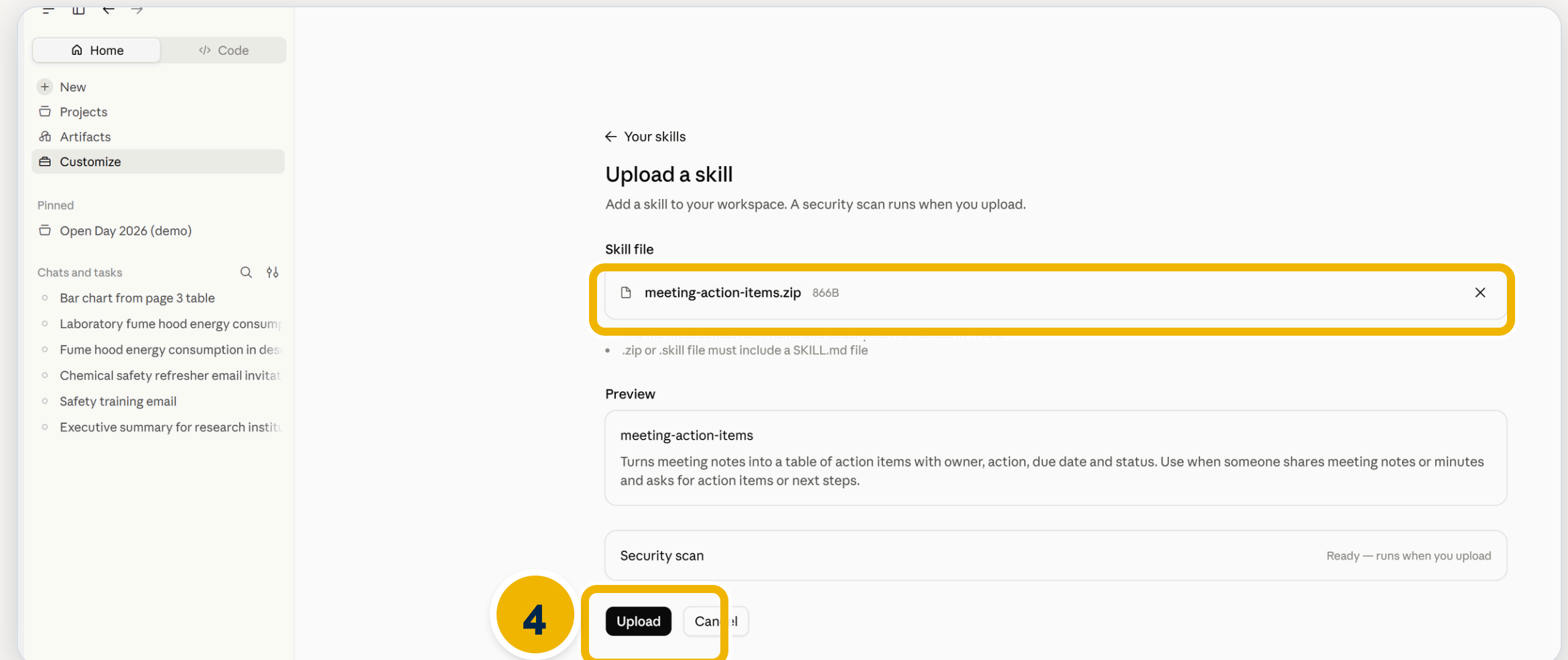
STEP 4 OF 7

Works on Free

Stuck? Go back up to the first "Your turn 2" slide, or ask a TA.

Pick the zip as it is, then Upload

Click the box, open **02-skill**, pick **meeting-action-items.zip** (a zip is one packed file; it has a zipper icon). Don't open it. Then **Upload**. A quick security scan runs.



Check it is switched on

The switch at the top right is blue.
 Below it you can read the skill: it's just instructions.

←
Your skills

meeting-action-items
New

by you · v1 · updated just now

5



Overview
Contents · 1

Meeting action items v1 · current

Download

Edit

SKILL.md

/SKILL.md

Meeting notes to an action-item table

When the user shares meeting notes (pasted or attached) and wants action items or next steps:

- Read all the notes first.
- Find every action: anything someone agreed to do, was asked to do, or must follow up. Ignore general discussion.
- Make one table with these columns, in this order: #, Owner, Action, Due date, Status.
 - Owner: the person's name exactly as written in the notes. If nobody was named, write "Unassigned".
 - Action: one short line that starts with a verb, for example "Book the main hall".
 - Due date: as written in the notes. If none was given, write "Not set".
 - Status: "Open", unless the notes say it is done ("Done") or stuck ("Blocked").
- Sort the rows by due date, earliest first. "Not set" goes last.
- Under the table, add a short list called "Questions to check": anything unclear, such as an action with no owner, two different dates, or a vague task.
- Never invent owners, dates or tasks that are not in the notes.
- Keep it short. Do not summarise the meeting unless asked.
- Reply in the same language as the notes.
- If the user asks for a spreadsheet, also create an Excel file with the same columns.

New chat, add the notes, ask

New, then + > Add files or photos > **meeting-notes-open-day**. Then type:

Turn these notes into action items.

meeting-notes-open-day.txt
14 lines
TXT

6 Turn these notes into action items.

Check skill for formatting action items table SKILL.md

#	Owner	Action	Due date	Status
1	Mr. Test Participant	Get 3 quotes from print shops for posters and banners	Thursday 9 Oct	Open
2	Ms. Example Name	Confirm main hall booking in writing	12 Oct	Open
3	Dr. Sample Person	Email department heads asking for 20 volunteers	15 Oct	Open
4	Ms. Example Name	Send budget sheet to finance	"end of next week" (17 Oct?)	Open
5	Mr. Demo Helper	Build open day web page	Not set	Done
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7	Facilities	Confirm where parking and direction signs can go	Not set	Blocked

Questions to check:

- Budget sheet due date is unclear — noted as "end of next week" (~17 Oct) but Mr. Test thought the 15th.
- Catering/dietary check has no owner — needs to be assigned at the next meeting.
- Parking/signage is blocked on Facilities, with no date for when they'll respond.
- If the main hall booking falls through, the fallback (atrium) has no named owner or deadline.

Check the skill was used

The grey line mentions **SKILL.md**: Claude opened your skill's instructions. The table has #, Owner, Action, Due date, Status, and there's a list called **Questions to check**.

meeting-notes-open-day.txt
14 lines
TXT

Turn these notes into action items.

7

check skill for formatting action items table SKILL.md

#	Owner	Action	Due date	Status
1	Mr. Test Participant	Get 3 quotes from print shops for posters and banners	Thursday 9 Oct	Open
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3	Dr. Sample Person	Email department heads asking for 20 volunteers	15 Oct	Open
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Questions to check:

- Budget sheet due date is unclear — noted as "end of next week" (~17 Oct) but Mr. Test thought the 15th.
- Catering/dietary check has no owner — needs to be assigned at the next meeting.
- Parking/signage is blocked on Facilities, with no date for when they'll respond.
- If the main hall booking falls through, the fallback (atrium) has no named owner or deadline.

— MAKE YOUR OWN

Ask Claude to write the skill for you

Customize > Skills > + Add > Create with Claude.
It fills in:

Let's create a skill together using your skill-creator skill. First ask me what the skill should do.

+

Chat

Cowork

Sonnet 5 Medium

↑

Claude asks three questions, drafts the skill, and shows **Save skill**. No files by hand.

Creating a custom skill

Let's create a skill together using your skill-creator skill. First ask me what the skill should do.

Reading skill-creator instructions before starting

What should this skill do? A few things that'll help me get started:

- What should it enable Claude to do?** (e.g., transform a specific file type, follow a particular writing style, automate a recurring workflow)
- When should it trigger?** — what phrases or situations should call it up
- What should the output look like?**

Once I know the basic idea, I'll ask a few follow-up questions about edge cases and examples before we draft it.

It should write short announcements for staff events. Use it when I ask for an event announcement. Output: a subject line, then under 100 words with the date, time, place, one line on why to come, and how to register. Friendly, clear tone. No need for more questions: please draft the skill now.

Ran 5 commands, created a file

No output-dir flag; it just writes into the current directory. Let me package from within the outputs directory instead.

Ran 3 commands, shared files

Here's the draft — it triggers on requests to announce a staff event, requires the event name, date, time, place, one reason to come, and RSVP info, and outputs a subject line plus a sub-100-word friendly body (see the example inside the file).

Want to test it on a real event of yours, or tweak anything about the tone/format first?

staff-event-announcement Skill

Save skill

Write a message...

Claude is AI and can make mistakes. Please double-check responses.

Sonnet 5 Medium

staff-event-announcement / SKILL.md

1

Save skill

Description

Write short internal announcements for staff/team events (office parties, town halls, trainings, happy hours, offsites, etc). Use this whenever the user asks for an "event announcement," wants to "announce" an event to staff/the team/employees, or asks to write a message inviting coworkers to something. Always use this skill for that request rather than writing a generic announcement from scratch, even if the user gives only partial details (date, time, place) — ask for whatever's missing, then produce the announcement in the required format.

Staff Event Announcement

Write a short, friendly announcement inviting staff to an event.

Required inputs

Before writing, make sure you have:

- Event name/type** (e.g., "Q3 Town Hall", "Summer Picnic")
- Date**
- Time**
- Place** (physical location, building/room, or a virtual link)
- Why to come** — one compelling reason (what's special, who's speaking, what's fun about it)
- How to register** — a link, an email to RSVP to, a sign-up sheet, or "just show up"

If the user hasn't given you one or more of these, ask a single, direct question listing exactly what's missing. Don't proceed with placeholders like "[TBD]" or invented details.

Output format

Produce exactly this structure:

Subject: <short, inviting subject line – under 10 words>

<body – under 100 words total>

The body must:

- Open with a warm, brief hook (1 sentence) — not a dry "This is to inform you"
- State the date, time, and place clearly (can be its own line or woven into a sentence)
- Include exactly one line on why people should come (the hook)
- End with clear registration/RSVP instructions
- Use a friendly, clear, upbeat tone — like a helpful colleague, not corporate memo-

Claude asks, you answer in plain words

SKILL

Let's create a skill together using your skill-creator skill. First ask me what the skill should do.

Reading skill-creator instructions before starting >

What should this skill do? A few things that'll help me get started:

1. **What should it enable Claude to do?** (e.g., transform a specific file type, follow a particular writing style, automate a recurring workflow)
2. **When should it trigger?** — what phrases or situations should call it up
3. **What should the output look like?**

Once I know the basic idea, I'll ask a few follow-up questions about edge cases and examples before we draft it.



— IT FOLLOWS YOU

Make it once, use it everywhere

One skill in your Claude account

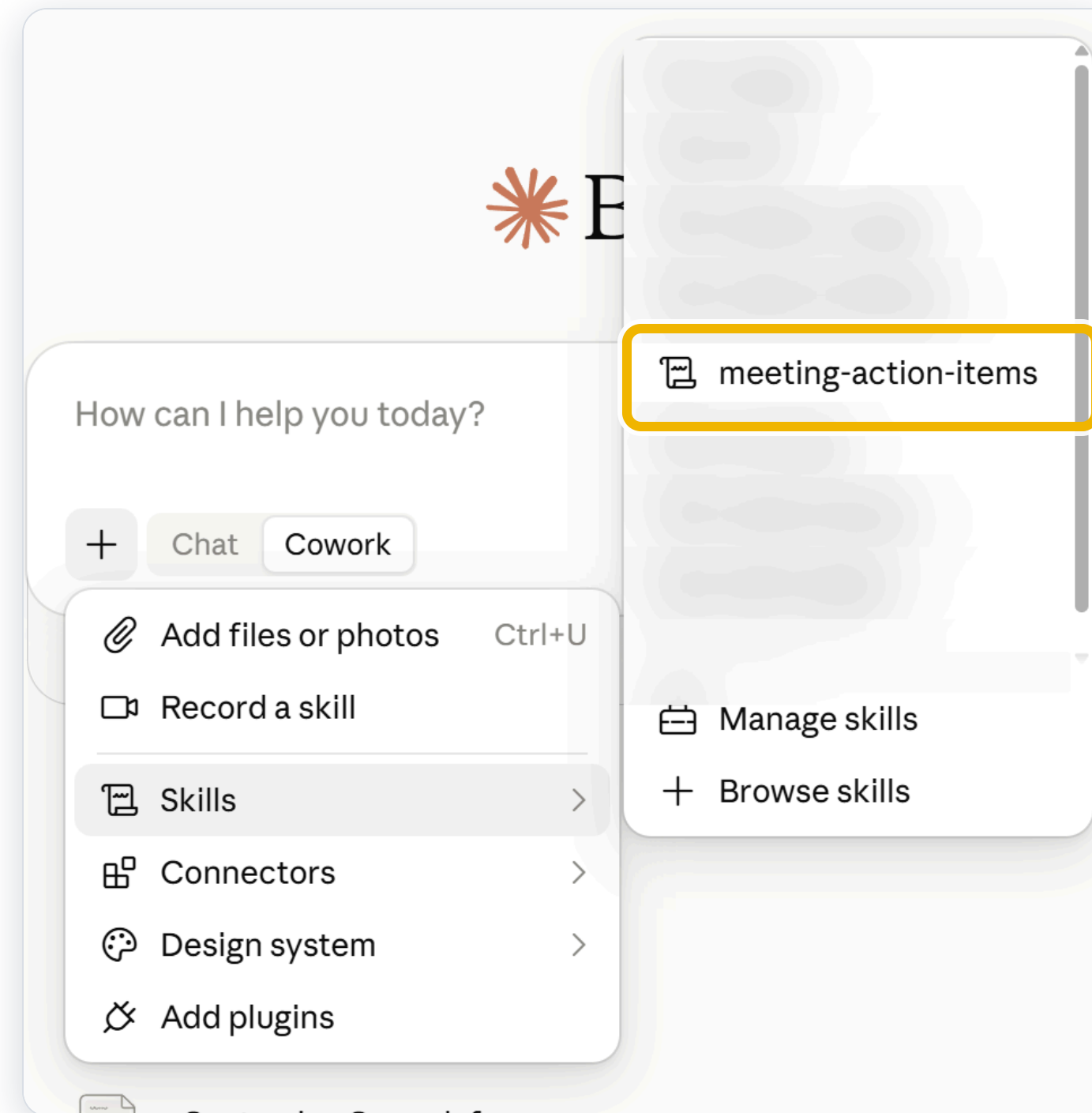


Chat

Cowork

Claude Code

The skill you just uploaded, in Cowork's + > **Skills** menu:



Claude works on your files

Cowork and Claude Code. Paid plans; on Free, watch and try the Chat version.

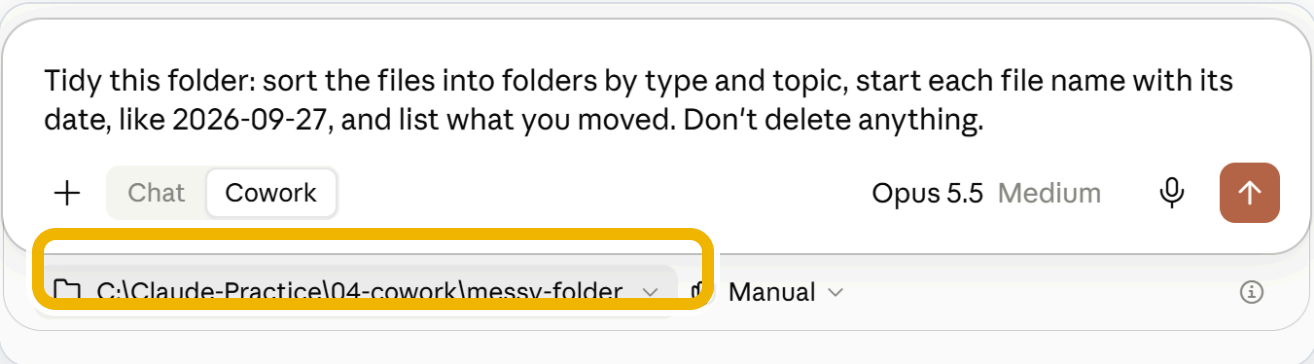
— WHICH ONE?

Chat, Cowork or Code?

	For	Pick it when
Chat	Questions, drafts, and summaries of files you attach	You'll copy the answer out yourself
Cowork	Handing Claude a job on a folder of files. Set to Manually approve, it checks with you before acting	You want a finished file or a tidy folder
Code	Claude works on a folder. In Manual mode it shows each exact change before it makes it. No coding needed: you ask in plain words	You want to see and approve every change

Start in Chat. Move to Cowork or Code when the job touches your files.

Tidy a messy folder



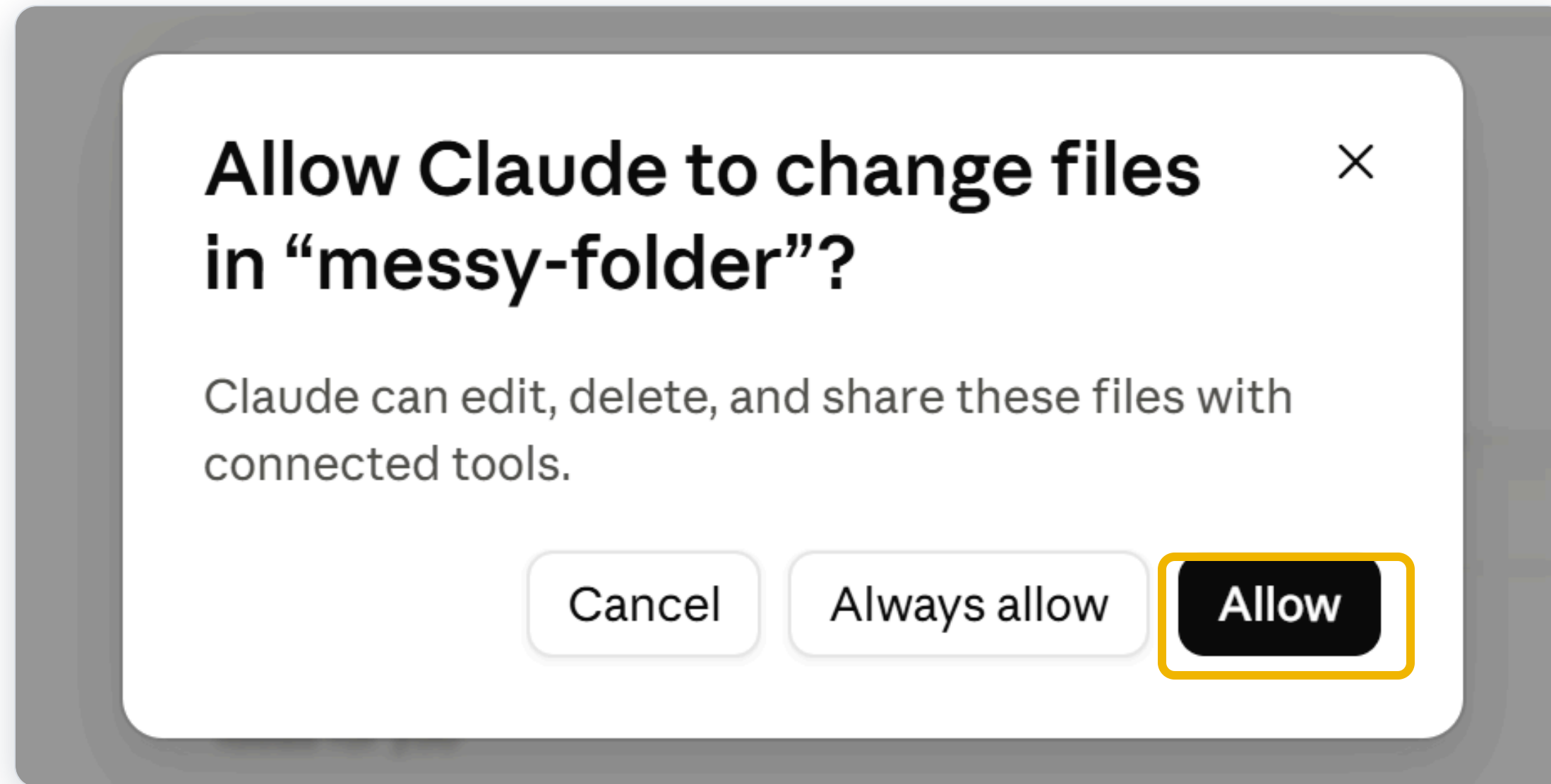
It asked about the folder, then worked. **12 files, 90 seconds, nothing deleted.**

Old name	New location and name
Copy of agenda.docx	Documents/Meetings/2026-09-21 Planning meeting agenda.docx
minutes 3 march.docx	Documents/Meetings/2026-03-03 Planning meeting minutes.docx
notes from tuesday.txt	Documents/Meetings/2026-09-16 Call notes - print shop banners.txt
final FINAL v3.docx	Documents/Programme/2026-09-14 Open Day programme v3.docx
speaker bios.docx	Documents/Programme/2026-09-10 Speaker bios.docx
Scan_0034.pdf	Documents/Letters/2026-09-05 Outreach Office letter - display stands.pdf
New Text Document.txt	Documents/Notes/2026-09-18 Reminder - Facilities parking signs.txt
photo list.txt	Documents/Notes/2026-09-12 Photo shot list.txt
budget draft (2).xlsx	Spreadsheets/Budget/2026-09-08 Open Day budget draft.xlsx
open day poster.png	Images/Posters/2026-09-15 Open Day poster draft.png
IMG_2231.png	Images/Photos/2026-09-11 Main hall photo.png
receipt-taxi.png	Images/Receipts/2026-09-09 Taxi receipt.png

Where a file had more than one date, I used the date it was written, updated or taken. For example, the letter gets 2026-09-05, not the 2 November event date it mentions.

2 minutes ago

It asks before touching the folder



You choose how often it checks with you

How can I help you today?

+

Chat

Cowork

Opus 5.5 Medium

🎤

📁 Project or folder ▾

👤 Manual ▾

ⓘ

Ideas for you

🌅 Send me a daily

👤 Manually approve ✓

🔗 Automatically approve

⚠ Skip all approvals

The result: a list of what moved

Old name	New location and name
Copy of agenda.docx	Documents/Meetings/2026-09-21 Planning meeting agenda.docx
minutes 3 march.docx	Documents/Meetings/2026-03-03 Planning meeting minutes.docx
notes from tuesday.txt	Documents/Meetings/2026-09-16 Call notes - print shop banners.txt
final FINAL v3.docx	Documents/Programme/2026-09-14 Open Day programme v3.docx
speaker bios.docx	Documents/Programme/2026-09-10 Speaker bios.docx
Scan_0034.pdf	Documents/Letters/2026-09-05 Outreach Office letter - display stands.pdf
New Text Document.txt	Documents/Notes/2026-09-18 Reminder - Facilities parking signs.txt
photo list.txt	Documents/Notes/2026-09-12 Photo shot list.txt
budget draft (2).xlsx	Spreadsheets/Budget/2026-09-08 Open Day budget draft.xlsx
open day poster.png	Images/Posters/2026-09-15 Open Day poster draft.png
IMG_2231.png	Images/Photos/2026-09-11 Main hall photo.png
receipt-taxi.png	Images/Receipts/2026-09-09 Taxi receipt.png

Where a file had more than one date, I used the date it was written, updated or taken. For example, the letter gets 2026-09-05, not the 2 November event date it mentions.

    2 minutes ago

10 supplier price quotes → one spreadsheet

Read the 10 supplier price quotes in the quotes folder and make one Excel sheet: supplier, item, quantity, unit price, total, delivery days. Highlight the cheapest for each item.

55 seconds. All seven "cheapest" picks checked and correct. It also flagged a trade-off: the cheapest chairs take 12 days.

There are 19 quote lines from the 10 suppliers, sorted by item. The total column multiplies quantity by unit price, and every total matches what the suppliers wrote. For each item, the row with the lowest unit price is shaded green, and the shading updates if you change a price.

Item	Cheapest supplier	Unit price	Total	Delivery
Banner printing (4)	Placeholder Print House	SAR 390.00	SAR 1,560	8 days
Coffee service (300 people)	Demo Catering Co.	SAR 16.50	SAR 4,950	5 days
Direction signs (30)	Example Event Rentals	SAR 49.00	SAR 1,470	6 days
Display tables (20)	Test Hospitality Group	SAR 140.00	SAR 2,800	7 days
Folding chairs (200)	Example Trading Est.	SAR 35.00	SAR 7,000	12 days
Name badges (500)	Sample Signs & Badges	SAR 1.90	SAR 950	8 days
Photo booth (1 day)	Mock Media Services	SAR 2,500.00	SAR 2,500	10 days

- **Folding chairs:** the cheapest option takes 12 days to deliver. Sample Events LLC delivers in 4 days, but costs SAR 1,200 more.
- **Photo booth:** only one supplier quoted, so there was nothing to compare.

All prices exclude VAT.

just now

— CLAUDE CODE

Claude working directly on the files in a folder on your computer

You do **not** need to write code. You ask in plain words.

The Code tab

Inside Claude Desktop. Nothing extra to install. We use this today.

The text window

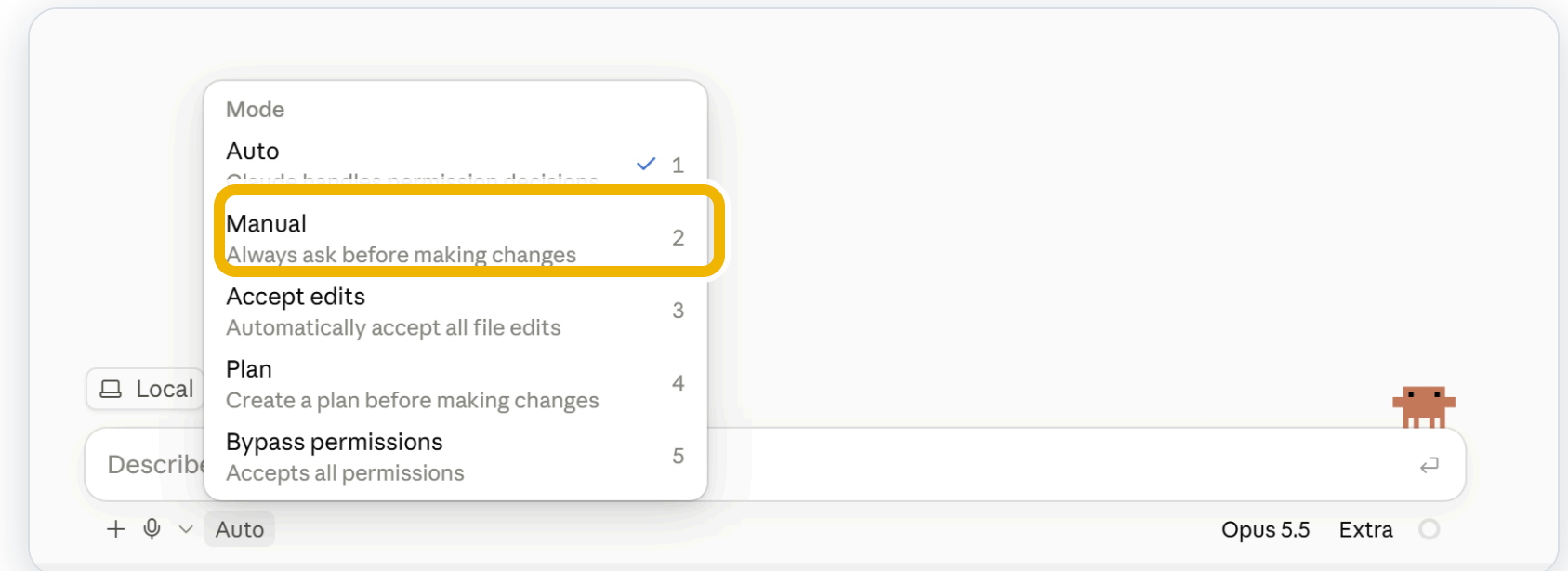
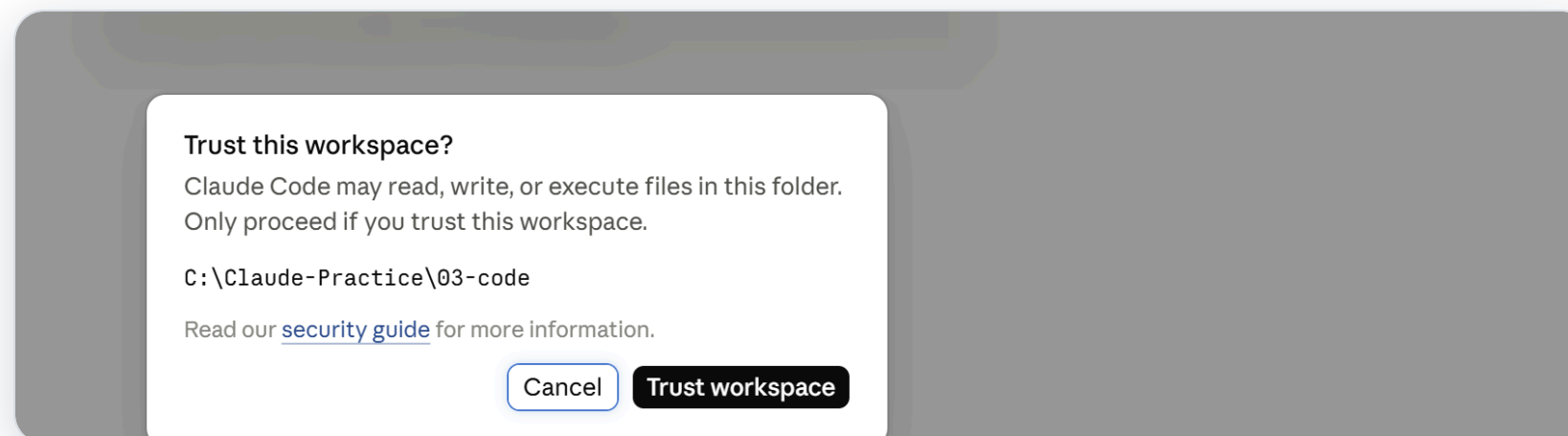
The same thing in PowerShell or Terminal: the text window where you type commands. Optional.

— THE CODE TAB

Pick a folder, pick a mode



- 1 **Local**: files on this computer
- 2 **the folder**: the practice folder
- 3 **the mode**: how often Claude stops to ask you first



Choose **Manual**: "**Always ask before making changes**". Only Manual asks before every change. Leave Auto and Bypass alone for now.

A new folder starts in Auto, so switch it to Manual. "Workspace" means the folder you picked. Your list may be shorter than this one.

Sort a list into files

Put the registrations into separate files, one per department, in a new folder called by-department. Then write a short summary with the number per department and the top 3.

In Manual it asked **3 times**. Chat can't do this: it made a new folder and 9 files.

Put the registrations into separate files, one per department, in a new folder called by-department. Then write a short summary with the number per department and the top 3.

Ran 3 commands, read a file, created a file +28-0 >

Split done (120 rows → 8 files, all accounted for). Now writing the summary file.

Created summary.md +23-0 >

I split the 120 registrations into 8 department files in [by-department/](#), one CSV per department. Each file keeps the original header, and the row counts add back up to 120.

Department	Registrations
Department of Imaginary Studies	24
Centre for Example Research	21
Office of Sample Services	19
Fictional Finance Office	14
Placeholder Engineering Unit	12
Test Facilities Management	11
Mock Student Affairs	10
Demo Library Services	9

Top 3:

- 1. Department of Imaginary Studies: 24 (20%)
- 2. Centre for Example Research: 21 (17.5%)
- 3. Office of Sample Services: 19 (about 16%)

Together, those three make up just over half of all registrations (64 of 120).

The file names are the department names in lowercase with hyphens, e.g. `department-of-imaginary-studies.csv`. I also saved this summary in the same folder as [summary.md](#). The original `registrations.csv` hasn't been changed.



A "may I do this?" question

Allow Claude to run Check Python availability and list old drafts?

Check Python availability and list old drafts

python --version; Get-ChildItem old-drafts | Select-Object Name

Deny 1 Esc

Always allow 2 Ctrl Shift ↵

Allow once 3 Ctrl ↵

Type / for commands

Read: what does it want to do? Then **Allow once**, or **Deny**.

The result

Put the registrations into separate files, one per department, in a new folder called by-department. Then write a short summary with the number per department and the top 3.

Ran 3 commands, read a file, created a file +28-0 >

Split done (120 rows → 8 files, all accounted for). Now writing the summary file.

Created summary.md +23-0 >

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Your turn 3

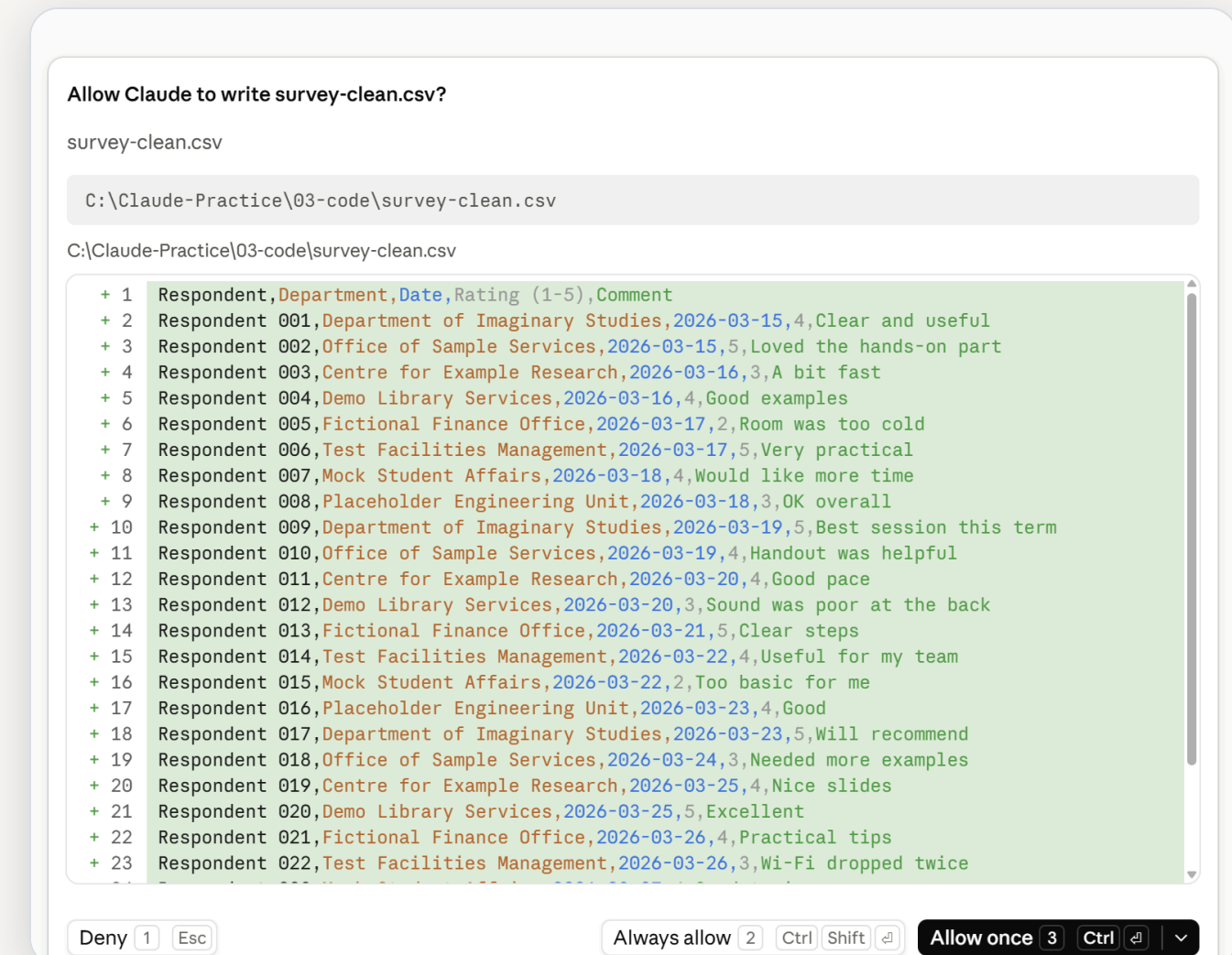
• 9 minutes

Needs a paid plan

Free: do it in Chat

Answer Claude's "may I do this?" questions

- 1 Click the **Code** tab.
- 2 **Local**, then the folder button > **Open folder...** > Claude-Practice > **03-code**. Then **Trust workspace**.
- 3 Mode selector (next to send): choose **Manual** ("Always ask before making changes"). Leave Auto and Bypass alone: they skip the questions.
- 4 Paste the Step 4 text from **text-to-copy.txt** and send.
- 5 Read each question: which file, what change? Green lines will be added. If it looks right: **Allow once**.
- 6 Paste the Step 6 text ("Also delete the old-drafts folder.") and send. Read the question, then click **Deny**.



Stuck? No Code tab or it asks to upgrade: you're on Free. Attach **survey-results** in Chat and paste the Step 4 (Free plan) text from **text-to-copy.txt**. No **Manual** in the list: choose Local (this computer), not Cloud (Anthropic's computers). It deleted without asking: the mode wasn't Manual; copy 03-code again from the zip.

If your screen looks different: you may see three, four or five modes. A Mac may offer "command line developer tools": click Not Now.

Your turn 3

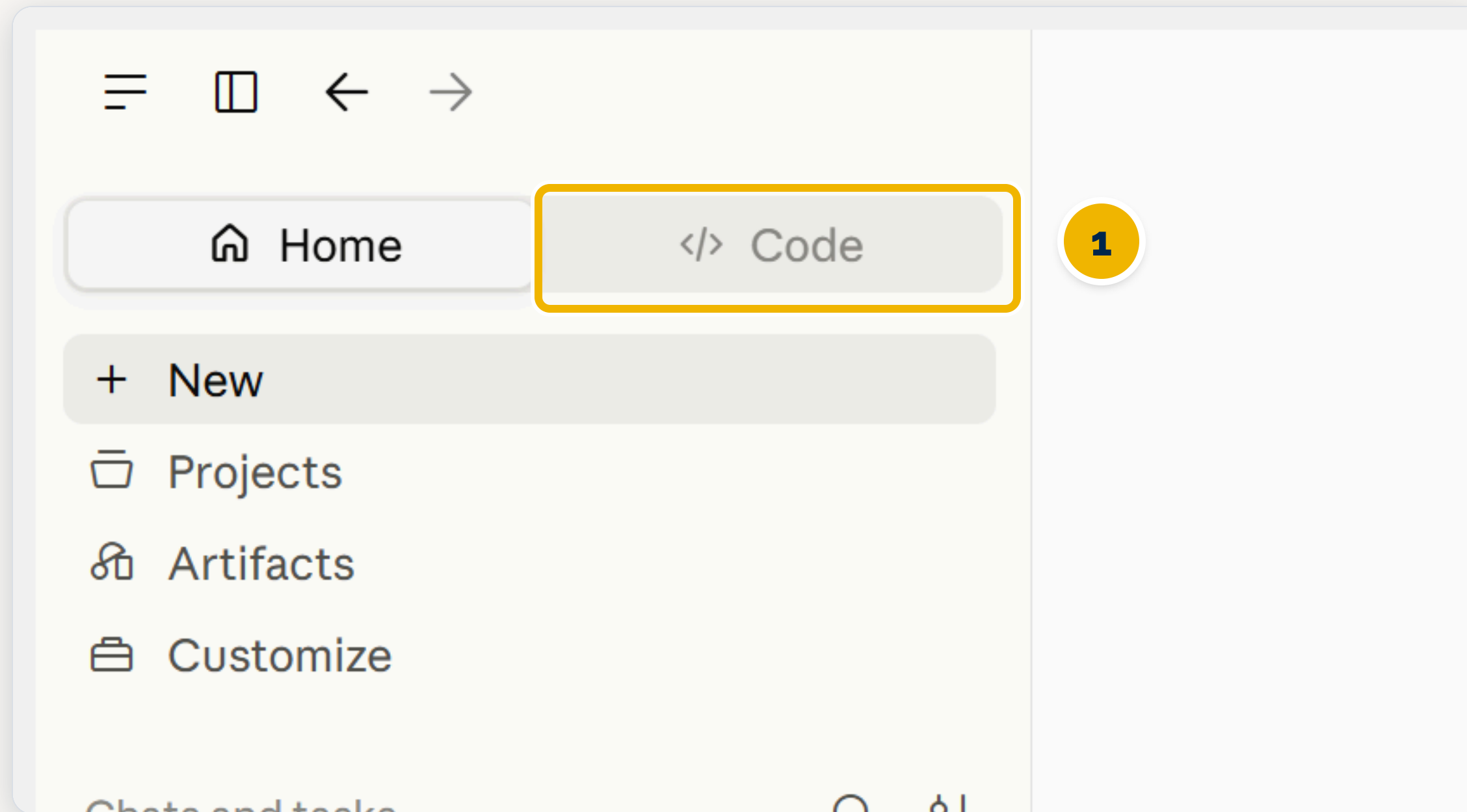
STEP 1 OF 6

Needs a paid plan

Free: do it in Chat

Stuck? Go back up to the first "Your turn 3" slide, or ask a TA.

Click the Code tab



On Free, a note says "Upgrade to Pro or Max to use Claude Code on the desktop." Do the Chat version instead.

Your turn 3

STEP 2 OF 6

Needs a paid plan

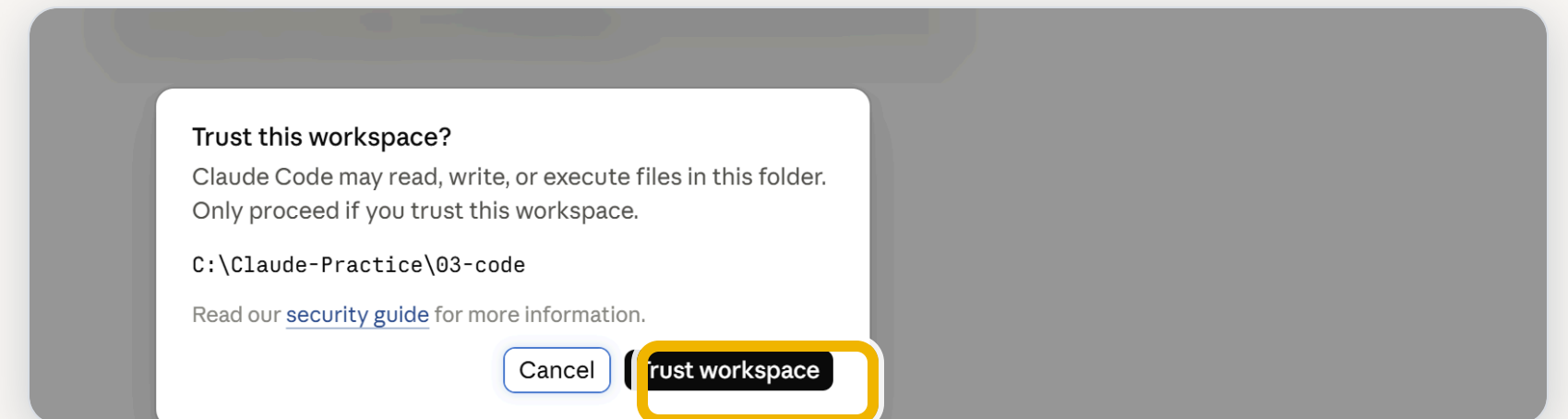
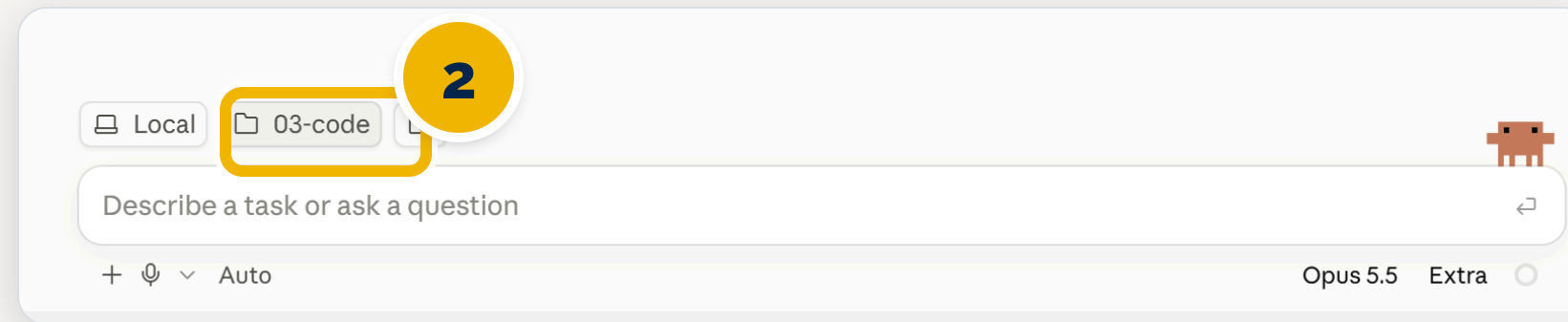
Free: do it in Chat

Stuck? Go back up to the first "Your turn 3" slide, or ask a TA.

Pick the 03-code folder, then trust it

Local, then the folder button, **Open folder...**, open Claude-Practice and choose **03-code** itself (not Documents).

"Trust this workspace?" It's your practice folder: **Trust workspace.**



Your turn 3

STEP 3 OF 6

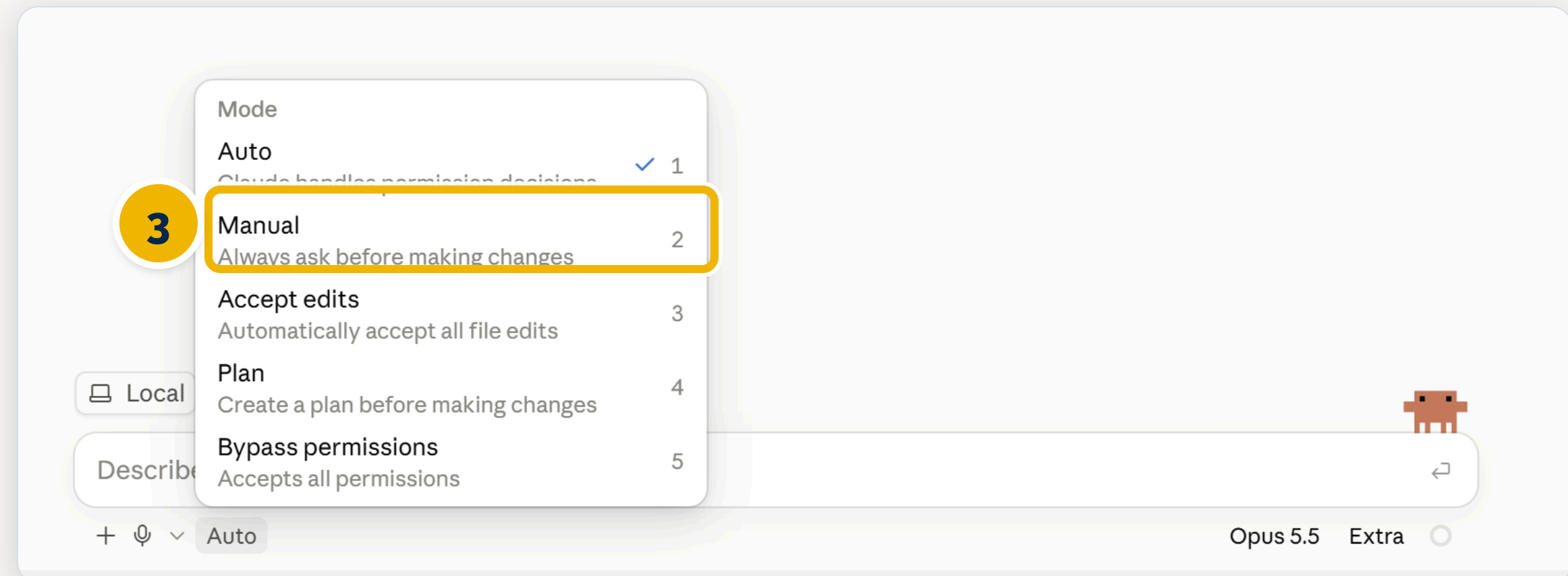
Needs a paid plan

Free: do it in Chat

Stuck? Go back up to the first "Your turn 3" slide, or ask a TA.

Choose Manual

Click the mode next to the send button. Choose **Manual: "Always ask before making changes"**. Leave Auto and Bypass alone: they skip the questions.



Your turn 3

STEP 4 OF 6

Needs a paid plan

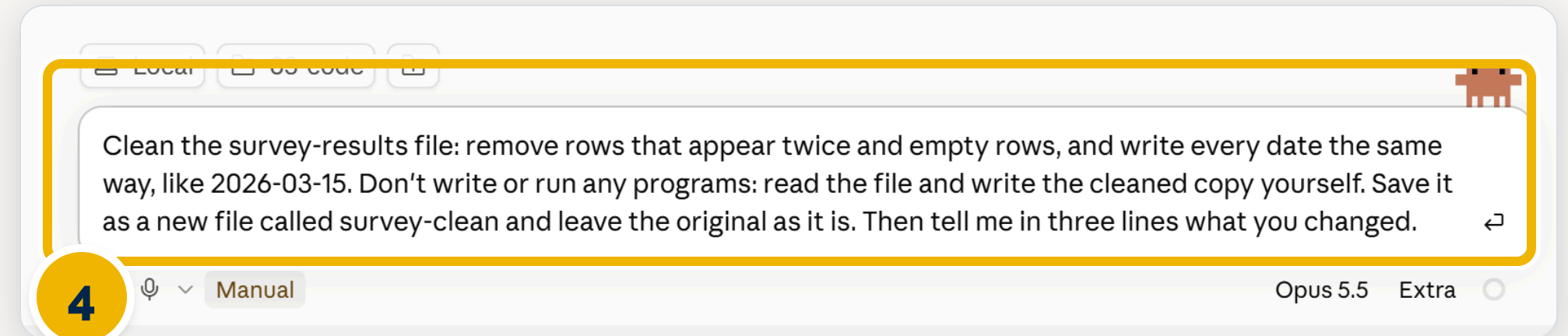
Free: do it in Chat

Stuck? Go back up to the first "Your turn 3" slide, or ask a TA.

Paste the request and send

Clean the survey-results file: remove rows that appear twice and empty rows, and write every date the same way, like 2026-03-15. Don't write or run any programs: read the file and write the cleaned copy yourself. Save it as a new file called survey-clean and leave the original as it is. Then tell me in three lines what you changed.

survey-results is a simple spreadsheet file (.csv). It opens in Excel.



Your turn 3

STEP 5 OF 6

Needs a paid plan

Free: do it in Chat

Stuck? Go back up to the first "Your turn 3" slide, or ask a TA.

Read it, then Allow once

Which file? **survey-clean.csv**, a new file. What change? The green lines will be added. Is that what you asked? Yes: **Allow once**.

A command (a line of computer text)? Check the file name and Claude's one-line reason. Not sure: **Deny** and ask "Explain in plain words what that would do."

Allow Claude to write survey-clean.csv?

survey-clean.csv

C:\Claude-Practice\03-code\survey-clean.csv

C:\Claude-Practice\03-code\survey-clean.csv

+ 1

Respondent,Department,Date,Rating (1-5),Comment

+ 2

Respondent 001,Department of Imaginary Studies,2026-03-15,4,Clear and useful

+ 3

Respondent 002,Office of Sample Services,2026-03-15,5,Loved the hands-on part

+ 4

Respondent 003,Centre for Example Research,2026-03-16,3,A bit fast

+ 5

Respondent 004,Demo Library Services,2026-03-16,4,Good examples

+ 6

Respondent 005,Fictional Finance Office,2026-03-17,2,Room was too cold

+ 7

Respondent 006,Test Facilities Management,2026-03-17,5,Very practical

+ 8

Respondent 007,Mock Student Affairs,2026-03-18,4,Would like more time

+ 9

Respondent 008,Placeholder Engineering Unit,2026-03-18,3,OK overall

+ 10

Respondent 009,Department of Imaginary Studies,2026-03-19,5,Best session this term

+ 11

Respondent 010,Office of Sample Services,2026-03-19,4,Handout was helpful

+ 12

Respondent 011,Centre for Example Research,2026-03-20,4,Good pace

+ 13

Respondent 012,Demo Library Services,2026-03-20,3,Sound was poor at the back

+ 14

Respondent 013,Fictional Finance Office,2026-03-21,5,Clear steps

+ 15

Respondent 014,Test Facilities Management,2026-03-22,4,Useful for my team

+ 16

Respondent 015,Mock Student Affairs,2026-03-22,2,Too basic for me

+ 17

Respondent 016,Placeholder Engineering Unit,2026-03-23,4,Good

+ 18

Respondent 017,Department of Imaginary Studies,2026-03-23,5,Will recommend

+ 19

Respondent 018,Office of Sample Services,2026-03-24,3,Needed more examples

+ 20

Respondent 019,Centre for Example Research,2026-03-25,4,Nice slides

+ 21

Respondent 020,Demo Library Services,2026-03-25,5,Excellent

+ 22

Respondent 021,Fictional Finance Office,2026-03-26,4,Practical tips

+ 23

Respondent 022,Test Facilities Management,2026-03-26,3,Wi-Fi dropped twice

Deny 1 Esc

Always allow 2 Ctrl

5

Allow once 3 Ctrl

Your turn 3

STEP 6 OF 6

Needs a paid plan

Free: do it in Chat

Stuck? Go back up to the first "Your turn 3" slide, or ask a TA.

Practise saying no: Deny

Paste the Step 6 text (*"Also delete the old-drafts folder."*).
Claude asks:

Allow Claude to run Move the old-drafts folder to the Recycle Bin and confirm it's gone?

Move the old-drafts folder to the Recycle Bin and confirm it's gone

```
Add-Type -AssemblyName Microsoft.VisualBasic;
[Microsoft.VisualBasic.FileIO.FileSystem]::DeleteDirectory("C:\Claude-Practice\03-code\old-drafts",
'OnlyErrorDialogs', 'SendToRecycleBin'); Test-Path "C:\Claude-Practice\03-code\old-drafts"
```

Deny 1

6

Always allow 2 Ctrl Shift

Allow once 3 Ctrl

After **Deny**, it stops. The folder stays.

Also delete the old-drafts folder.

Found files, ran a command >

The folder has three small text files and nothing hidden. I'll send it to the Recycle Bin rather than delete it permanently, so you can get it back if you need to.

Failed to move the old-drafts folder to the Recycle Bin and confirm it's gone >

✱

If you have time: *Don't delete it. Move it into a folder called archive instead.*

— THE OTHER WAY IN

The same thing, in a text window

- 1 Open **PowerShell** (Windows) or **Terminal** (Mac): the text window where you type commands.
- 2 Type `cd Documents\Claude-Practice` and press Enter (`cd` means "go to this folder"; on a Mac use `/`). Then type `claude` and press Enter.
- 3 It asks if you trust this folder: yes.
- 4 Press `Shift` + `Tab` until it says **manual mode on**.

```
Claude Code v2.1.283  
C:\Claude-Practice\06-terminal
```

```
> Try "how do I log an error?"
```

```
|| manual mode on
```

Rebuilt from the text of a real run, 27 September 2026. It starts in "auto mode on"; one Shift+Tab gives manual.

First time in a folder: a safety check

Accessing workspace:

C:\Claude-Practice\06-terminal

Quick safety check: Is this a project you created or one you trust? (Like your own code, a well-known open source project, or work from your team). If not, take a moment to review what's in this folder first.

Claude Code'll be able to read, edit, and execute files here.

› No, exit

Yes, I trust this folder

Enter to confirm · Esc to cancel

Rebuilt from the text of a real run, 27 September 2026. It's your practice folder: press the down arrow to pick "Yes, I trust this folder", then Enter.

"May I do this?" in the text window

› Make a copy of survey-results.csv called backup.csv.

PowerShell command

```
| if (Test-Path backup.csv) { "backup.csv already exists" } else { Copy-Item  
| survey-results.csv backup.csv; Get-Item survey-results.csv, backup.csv | Select-Object Name,  
| Length }
```

Copy survey-results.csv to backup.csv if backup.csv doesn't exist

Do you want to proceed?

- › 1. Yes
- 2. No

Esc to cancel · Tab to amend

Rebuilt from the text of a real run. The green line is Claude's reason. After Esc: "Interrupted · What should Claude do instead?"

— IT FOLLOWS YOU

Your meeting-notes skill is here too

In the text window, type `/skills` and press Enter to see your skills.

claude.ai sync means skills from your Claude account, like the one you uploaded in Your turn 2.

Skills

17 skills · enter/space to cycle, / to search, t to sort, Esc to close

✓ docx	claude.ai sync
✓ meeting-action-items	claude.ai sync
✓ pdf	claude.ai sync
✓ pptx	claude.ai sync
✓ xlsx	claude.ai sync

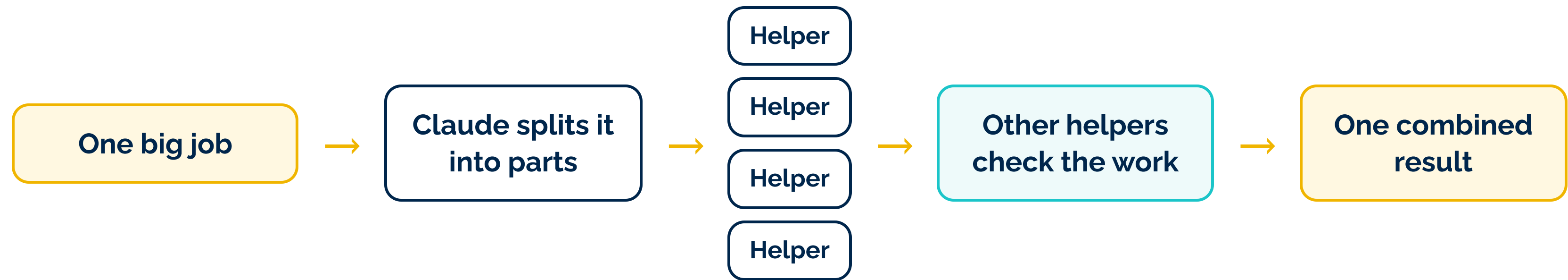
Rebuilt from the text of a real run; only skills from Anthropic and the demo skill are shown.

Workflows: many helpers at once

"Workflow" here means a Claude feature, not the everyday word.

WHAT A WORKFLOW IS

Split the job, work at the same time, check, combine



A **helper** is a Claude that works on one part of the job. They work at the same time.

— WHEN

Worth it, or not?

Worth it

- Big jobs: many files, many sources
- Repetitive jobs: the same steps 30 times
- High-stakes jobs: you want claims checked

Not worth it

- Quick questions
- A single email or summary
- Anything you need in one minute

It uses much more of your plan than a normal request, so you may reach your limit and have to wait. Our research run used about a quarter of what one Team account can use in 5 hours.

What the helpers found

109

helpers

~12 min

in the background

27

sources read

22 of 25

checked claims held up; 3
were refuted

Finished 6 background commands (2 failed), ran 4 commands (1 failed) >

Cutting energy use in university research labs

The biggest, best-proven savings come from **lab ventilation** (fume hoods and how many times per hour the room air is replaced, called air changes per hour or ACH) and **ultra-low temperature (ULT) freezer set points**. Lighting upgrades are reliable but smaller. Equipment scheduling is sensible, but the evidence for how much it saves is the weakest of the four.

The workflow searched 6 angles, fetched 27 sources and pulled 124 claims from them. It checked the top 25 claims with three independent checkers each: 22 held up and 3 were refuted. It skipped the other ~100 because of its budget.

Measure

Typical documented savings

Evidence

Background tasks: each line is one helper

Energy use reduction in research labs05-research

/deep-research

What are proven, practical ways to cut energy use in university research labs? Cover fume hoods, freezers, equipment scheduling and lighting, and include sources.

Auto mode is now Claude Code's default permission mode.

Auto mode lets Claude handle permission prompts automatically. Claude checks each tool call for risky actions and prompt injection before executing, runs the ones it assesses as lower-risk, and blocks the rest.

code.claude.com/docs/en/permission-modes

deep-research

Workflow34 agents3m 07s

The deep-research workflow is now running in the background. It will:

1. Split your question into 5 search angles: fume hoods, freezers, equipment scheduling, lighting, and lab-wide programs.

2. Run parallel web searches and fetch the top ~15 sources.

3. Pull out specific claims that can be checked, such as "a Sash-closing campaign saves X kWh," and have three independent checkers try to disprove each one. A claim is dropped if two of the three refute it.

4. Write a ranked report with citations.

I'll pass on the findings once it finishes. You can follow progress with /workflows if you want.

1 background command completed, 1 failed

1 running task

Type / for commands

Opus 5.5Extra

Background tasks

deep-research

Workflow3m 07s

34 agents1.8M tokens

Deep research harness — fan-out web searches, fetch sources, adversarially verify claims, synthesize a cited report.

Phases

Scope

1/1

Agent	Model	Tokens	Time
✓ scope	Opus 5.5	44.8k	13s

Search

6/6

Agent	Model	Tokens	Time
✓ search:Fume hoods: sash management and VA...	Opus 5.5	53.6k	44s
✓ search:Ultra-low temperature (ULT) freezers: -...	Opus 5.5	54.5k	42s
✓ search:Equipment scheduling, shutdown and ...	Opus 5.5	62.3k	1m 28s
✓ search:Lighting and occupancy controls in lab...	Opus 5.5	53.6k	46s
✓ search:Authoritative frameworks and green la...	Opus 5.5	53.4k	40s
✓ search:Skeptical view: safety tradeoffs, behav...	Opus 5.5	55.2k	45s

Fetch

26/27

Agent	Model	Tokens	Time
✓ fetch:betterbuildingsssolutioncenter.energy.gov	Opus 5.5	50.7k	1m 36s
✓ fetch:smartlabs.i2sl.org	Opus 5.5	49.2k	35s
✓ fetch:pubs.rsc.org	Opus 5.5	62.5k	1m 52s
✓ fetch:labmanager.com	Opus 5.5	48.5k	29s
✓ fetch:ucl.ac.uk	Opus 5.5	54.1k	56s
✓ fetch:smartlabs.i2sl.org	Opus 5.5	48.4k	25s
✓ fetch:colorado.edu	Opus 5.5	49k	34s
✓ fetch:inderscience.com	Opus 5.5	48.3k	31s
✓ fetch:ncbi.nlm.nih.gov	Opus 5.5	60.8k	2m 06s
✓ fetch:mygreenlab.org	Opus 5.5	48.4k	30s
✓ fetch:colorado.edu	Opus 5.5	48.5k	28s
✓ fetch:energy.gov	Opus 5.5	47.8k	22s
✓ fetch:sustainable.harvard.edu	Opus 5.5	57.4k	43s
✓ fetch:energy.gov	Opus 5.5	48.4k	29s
✓ fetch:www1.eere.energy.gov	Opus 5.5	53.7k	40s
✓ fetch:nature.com	Opus 5.5	50.3k	42s
✓ fetch:frontiersin.org	Opus 5.5	48.6k	35s
✓ fetch:ncbi.nlm.nih.gov	Opus 5.5	50.1k	46s
✓ fetch:web.mit.edu	Opus 5.5	66.3k	45s
✓ fetch:sid.siemens.com	Opus 5.5	63.9k	35s
✓ fetch:tandfonline.com	Opus 5.5	48.9k	34s
✓ fetch:mygreenlab.org	Opus 5.5	48.9k	29s
✓ fetch:smartlabs.i2sl.org	Opus 5.5	50.8k	36s
✓ fetch:smartlabs.i2sl.org	Opus 5.5	48.6k	31s
✓ fetch:frontiersin.org	Opus 5.5	49.3k	38s
✓ fetch:today.umd.edu	Opus 5.5	48.3k	25s

109 helpers, about 12 minutes

Background tasks

Running

deep-research

Workflow 9m 37s

109 agents 6M tokens

Deep research harness — fan-out web searches, fetch sources, adversarially verify claims, synthesize a cited report.

Phases

Scope

1/1

Agent

Model

Tokens

Time

The report, with how it was checked

uction in research labs ▾

05-research

/deep-research

What are proven, practical ways to cut energy use in university research labs? Cover fume hoods, freezers, equipment scheduling and lighting, and include sources.

Auto mode is now Claude Code's default permission mode.

Auto mode lets Claude handle permission prompts automatically. Claude checks each tool call for risky actions and prompt injection before executing, runs the ones it assesses as lower-risk, and blocks the rest.

code.claude.com/docs/en/permission-modes

Ran workflow deep-research

The deep-research workflow is now running in the background. It will:

1. Split your question into 5 search angles: fume hoods, freezers, equipment scheduling, lighting, and lab-wide programs.
2. Run parallel web searches and fetch the top ~15 sources.
3. Pull out specific claims that can be checked, such as "a Sash-closing campaign saves X kWh," and have three independent checkers try to disprove each one. A claim is dropped if two of the three refute it.
4. Write a ranked report with citations.

I'll pass on the findings once it finishes. You can follow progress with `/workflows` if you want.

Finished 6 background commands (2 failed), ran 4 commands (1 failed) >

Cutting energy use in university research labs

The biggest, best-proven savings come from **lab ventilation** (fume hoods and how many times per hour the room air is replaced, called air changes per hour or ACH) and **ultra-low temperature (ULT) freezer set points**. Lighting upgrades are reliable but smaller. Equipment scheduling is sensible, but the evidence for how much it saves is the weakest of the four.

The workflow searched 6 angles, fetched 27 sources and pulled 124 claims from them. It checked the top 25 claims with three independent checkers each: 22 held up and 3 were refuted. It skipped the other ~100 because of its budget.

Measure	Typical documented savings	Evidence
Shut the sash on variable air volume (VAV) hoods	~\$1,100-1,900 per hood per year	Strong
Lower ACH and reduce it further when the room is empty	\$50k-\$300k/yr per building	Strong
Whole-building "Smart Labs" retrofit (UC Irvine)	58-61% of building energy	Strong, but self-reported
ULT freezers at -70 °C instead of -80 °C	~12-37% per unit (2-7 kWh/day)	Strong
LEDs, then LEDs plus occupancy and daylight controls	~50%, up to ~80% of lighting energy	Moderate

Claims that were refuted are listed

reduction in research labs ▾ 05-research

- Most of those savings were estimated from published figures rather than metered. The per-item shutdown savings relied verification (see below), so no reliable figure exists yet for scheduling alone.

Claims that were refuted (don't cite these)

- DOE FEMP's "40% savings and longer equipment life" at -70 °C (0-3).
- "A ULT freezer uses about 20 kWh/day, like a household" (1-2).
- Iowa's per-item figures: about 23,400 kWh/yr for the autoclave and 8,300 kWh/yr for the biosafety cabinets (0-3).

Caveats

- **Most dollar figures are modeled or self-reported, and several are old.** Examples: Harvard used 2015 prices, and the DOE ventilation document and NIH study date from 2013.
- **Results depend on the site:** energy prices, climate, starting ACH and hood type. It is still unproven whether UC Irvine-style ACH cuts carry over to hot, humid climates where cooling and dehumidification dominate.
- **Behavior-based gains may fade.** A Toronto poster campaign lost its effect within months, and Harvard users disabled the automatic sash closers.
- **Any change to ACH or face velocity needs safety review first:** EHS sign-off and containment testing (ASHRAE 110, ANSI/AIHA Z9.5).
- **Not every sample is proven safe at -70 °C,** so check before moving sensitive stocks.

Sources

1. Harvard, *Shut the Sash* fume hood white paper: <https://sustainable.harvard.edu/wp-content/uploads/2023/09/FumeHoodWhitePaper-1.pdf>
2. Kongoletos et al., *Sci Rep* 11:21412 (2021), MIT sash alarms: <https://www.nature.com/articles/s41598-021-00772-y>
3. DOE Better Buildings Alliance, lab air change rate highlights: https://www1.eere.energy.gov/buildings/commercial/pdfs/bba_air_change_rates_highlights.pdf
4. I2SL, Emory occupancy sensor case study: <https://smartlabs.i2sl.org/cs-emory-occupancy-sensors.html>
5. DOE FEMP, UC Irvine Smart Labs success story: <https://www.energy.gov/cmei/femp/articles/university-california-irvine-smart-labs-initiative-success-story>
6. CU Boulder Green Labs, freezer energy efforts: <https://www.colorado.edu/ecenter/programs/cu-green-labs-program/freezers/freezer-energy-efforts>
7. NIH metered ULT freezer field study (Inderscience, 2013): <https://www.inderscience.com/info/inarticle.php?artid=50786>
8. Tran et al., *Frontiers in Sustainability* (2026), University of Iowa low-cost measures: <https://www.frontiersin.org/journals/sustainability/articles/10.3389/frsus.2026.1729128/full>
9. CU Boulder shared ULT freezer program: <https://www.colorado.edu/ecenter/programs/cu-green-labs-program/freezers/shared-ultra-low-temperature-freezer-program>
10. DOE FEMP, buying energy-efficient lab-grade refrigerators and freezers: <https://www.energy.gov/cmei/femp/purchasing-energy-efficient-laboratory-grade-refrigerators-and-freezers>

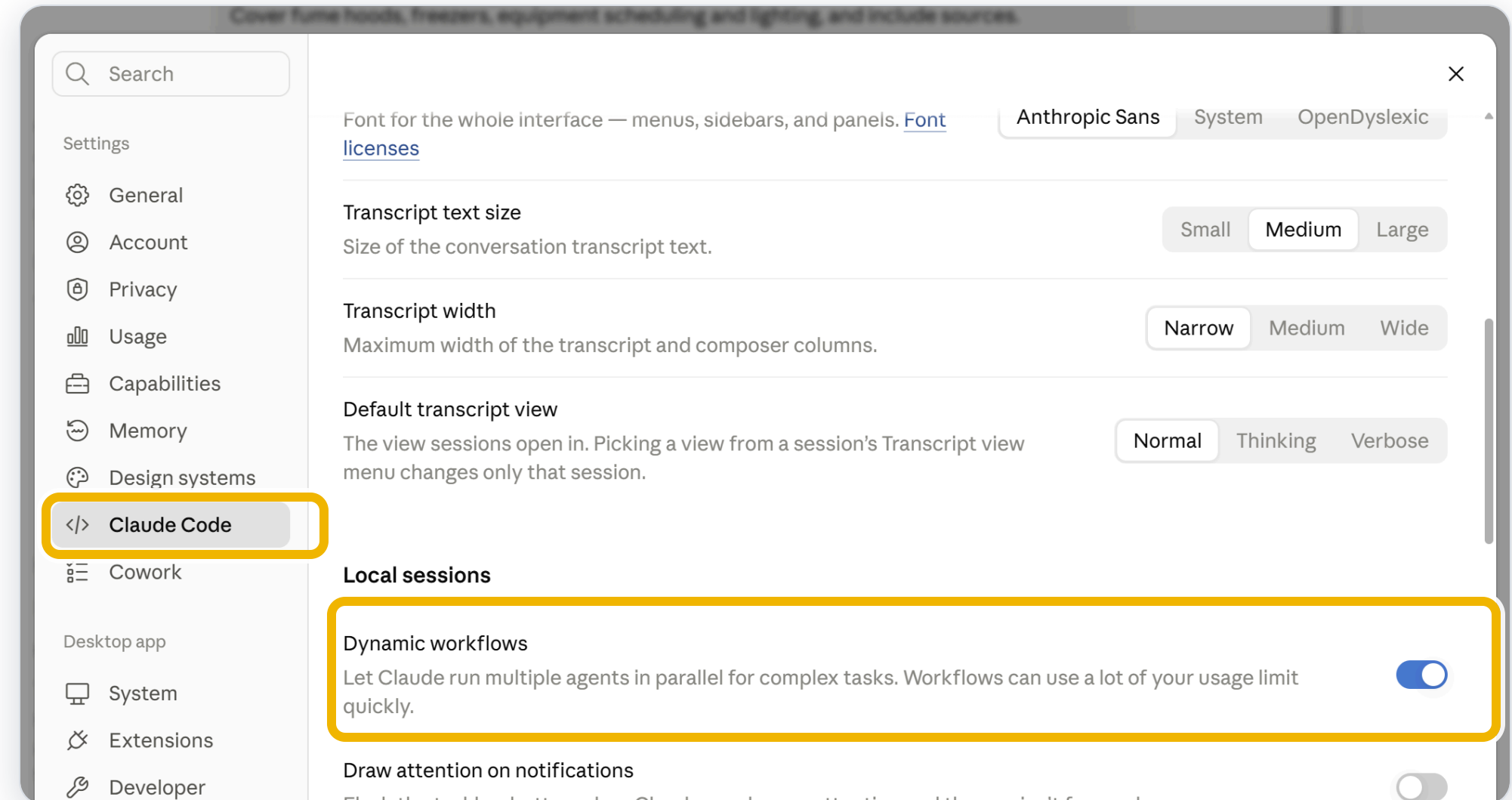
Want me to turn this into a shareable web page you can pass to facilities or EHS colleagues?

✱

— TRY IT LATER

Try it on a small job first

- 1 Paid plan, **Code** tab.
- 2 Pro plan: switch it on first. **Settings** > **Claude Code** > **Local sessions** > **Dynamic workflows**.
- 3 Type `/deep-research` , a space, and your question.
- 4 Start with a small question. Watch it in **Background tasks**.



Skills and workflows together

Together: a workflow where every helper follows your skill.

Use a workflow: give each of the 10 meeting-notes files in this folder to its own helper, and have every helper follow the meeting-action-items skill. Then combine all the tables into one.

Real run: **10 helpers, 30 seconds**, one table of 22 actions, plus "Questions to check".

Meeting	Owner	Action	Due
01 – Newsletter	Mr. Test Participant	Collect three staff stories	10 Oct
01 – Newsletter	Unassigned	Arrange the printing	Not set
02 – Budget review	Dr. Sample Person	Send Q3 figures to the committee	8 Oct
03 – Safety walk-round	Mr. Demo Helper	Fix the blocked fire exit sign	6 Oct
04 – New starters	Ms. Fictional Planner	Prepare welcome packs	12 Oct
...	...	...	...
10 – Annual report	Ms. Example Name	Collect section drafts	18 Oct

— SAFE USE

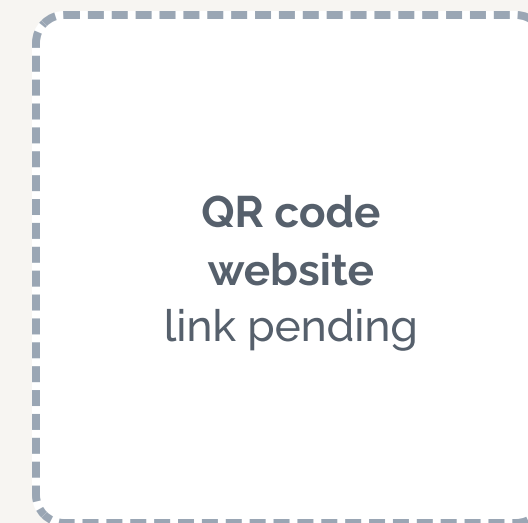
The five rules again

- 1 Never put in secrets or private data.
- 2 Know your account's rules, and KAUST's.
- 3 Practice folder only when Claude works on files.
- 4 Read every "may I do this?" before you say yes.
- 5 Check the answer before you use it.

You can now ...

- Get a useful answer: context, what you want back, changes.
- Let Claude work on a folder while you approve each change.
- Save a repeat task as a skill; know when a workflow is worth it.
- Keep private data out, and check what comes out.

Next: try one real (non-sensitive) task this week. The handout has every step.



Slides, handout, practice files

Questions?

QR code
feedback form
link pending

Ali Habibullah · KAUST Academy

Thank you. Please leave feedback before you go.